OCI Compute Instance Shape Change and Upgrade Methods

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OCI Cloud Compute Black Belt Team - EMEA

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Contents

OCI Compute Instance Shape Change and Upgrade Methods 1
Compute Instance Shape Upgrade Methods 3
1.1 Prerequisites and Considerations 3
How to Select Best Method for Changing/Upgrade Shape 4
In Place Change Shape 4
3.1 Pre-checks 4
3.2 In-place shape change (OCI Console) 6
3.3 Post-change validation 7
Custom Image Method for Linux OS Instance (Replace Boot Volume) 7
4.1 Pre-checks 8
4.2 Change Shape by Replacing Boot Volume 10
4.3 Post Activity Validation 14
Custom Image Method for Windows OS Instances 15
5.1 Preparation 16
5.2 Custom Image and New Instance Creation 18
5.3 Post Instance Creation Validation 20
Upgrading Existing Shapes to New Shapes with Acceleron 20
In Place Shape Upgrade Method 21
7.1 Pre-checks for In Place Upgrade 21
7.2 In-place shape change (OCI Console) 22
7.3 Post-change validation 23
Boot volume replacement (for Linux Instances) 24
7.4 Pre-checks 24
7.5 Change Shape by Replacing Boot Volume 27
7.6 Post Activity Validation 30

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Compute Instance Shape Upgrade Methods

This document provides additional technical detail to supplement the Oracle Support notes referenced below:

- **"OCI Compute - How to Update the Compute Instance Shape to/from Any Shape: Example VM.Standard.E5.Flex from a Platform Image that was Published Before the Shape was Released"**(Support KB155218) https://support.oracle.com/ic/builder/rt/customer_portal/live/webApps/customer-portal/?anchorId=&documentId=KB155218&page=sptemplate&sptemplate=km-article
- **"OCI Compute - How to Migrate from Older Fixed Shapes to Newer Flex Shapes"** (Support KB82990) https://support.oracle.com/support/?anchorId=&documentId=KB82990&page=sptemplate&sptemplate=km-article

This Youtube Video prepared by OCI Compute Black Belt team is useful to get more understanding of method described in the document : https://www.youtube.com/watch?v=mXr5sosWdvI

Before starting the activities, Customers are strongly encouraged to validate these methods in **test and non-production environments** first, and to confirm **verified backups** are in place before making any changes in production.

There are multiple methods to upgrade or change compute shapes (for example, moving from older fixed shapes to newer flex shapes). The best approach depends on **shape compatibility**, **regional availability**, **OS version/kernel**, and other platform constraints. Customers should review the options below, select the method that best fits their requirements, and execute the steps carefully.

1.1 Prerequisites and Considerations

1. **Unsupported Image Types and Features**

   OCI Marketplace images do not support upgrading and changing shapes with Custom Image Method for both Operating System types.

   DenseIO Shapes are excluded due to locally attached NVME SSDs.

   Shield Instances do not support In Place Shape Upgrade operation.

2. **Update OS and kernel**
   It is strongly recommended to update the operating system packages and kernel to the latest supported versions prior to the shape change. This reduces the risk of driver incompatibilities and unexpected boot issues when moving to newer shapes.

3. **Perform a validation reboot**
   Reboot the instance at least once before the upgrade activity to confirm the system can restart cleanly and that no existing issues would prevent the OS from booting.

4. **Ensure backups are in place**
   Take the necessary **boot volume** and **block volume** backups (and verify they are usable) prior to performing any shape change. <u>It is essential to take a manual backup of disks when Instance is in Shutdown State especially for Windows OS Instances with AD joined configurations.</u>

5. **Verify OS-specific configurations are preserved**
   Review and document any OS settings and customizations prior to the shape change to ensure they remain intact afterward. Examples include Active Directory domain join status, Group Policy settings, and NFS/CIFS mount configurations and others.

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6. **Operating system support limitations**
   Some of OS versions especially older kernels may not be supported on newer shapes or newer underlying virtualization/hardware. This can result in boot failures, missing drivers, or unsupported configurations after the shape change. It is very important to take backup and preserve disks before totally termination of old Instances.

7. **Capacity/availability constraints**
   The target shape may not have available capacity in a specific Availability Domain/Fault Domain, requiring an alternate AD/FD or a different shape selection.

8. **Troubleshooting**

   In case of any unforeseen issues such as Instance's unreachability after upgrade/change operation, Customers are advised to create local console connection and observe boot process. There are different methods are listed in below documentation.

   https://github.com/Olygo/OCI_Console-Connections

How to Select Best Method for Changing/Upgrade Shape

If Targeted shapes are available in Shape List → Use In Place Change Shape Method

If Targeted shapes are not available in Shape list and OS is Linux flavors → Use Custom Image Method for Linux OS Instance (Replace Boot Volume) Method

If Targeted shapes are not available in Shape list and OS is Windows Editions → Use Custom Image Method for Windows OS Instances Method

In Place Change Shape

This is a step-by-step guide to change an OCI Compute instance from one shape to another. We will use E5 and E6 Shapes in this example.

3.1 Pre-checks

- **Identify the current instance details**

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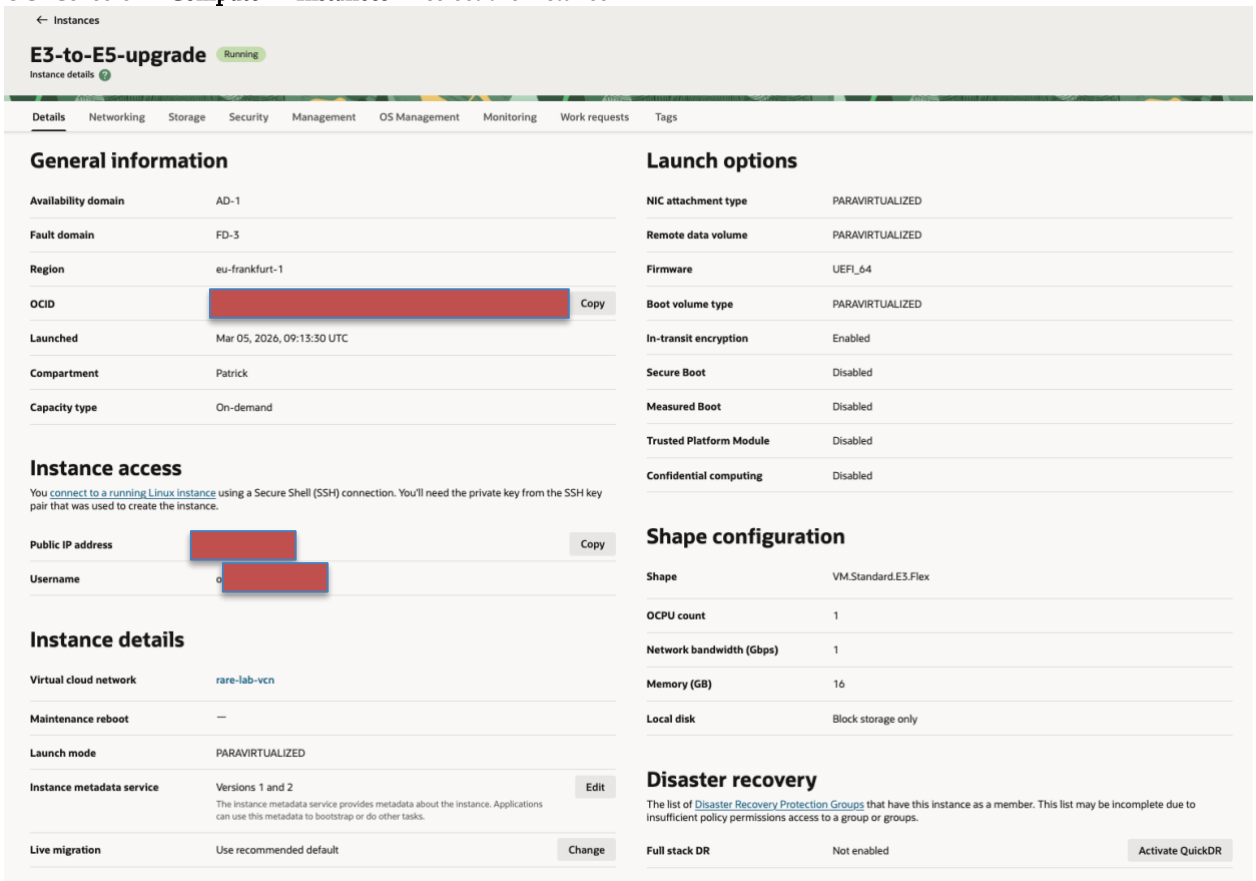
- OCI Console → **Compute** → **Instances** → select the instance



  - Current **shape**
  - **OCPUs / memory**
  - **Availability domain** (if applicable) and **fault domain**
  - **VNIC/subnet**, private IP, public IP (**ephemeral vs reserved**)
  - Attached **block volumes**, mount points, and any special boot configuration
- **Confirm New Shape availability and limit**
  - Console → **Governance & Administration** → **Limits, Quotas and Usage**
  - Validate you have sufficient **OCPUs** for the target E5/E6 shape in the correct **compartment/region/AD**.
  - Confirm the target shape is available in the instance's placement (some shapes are not available in every AD) using the OCI_ComputeCapacityReport.

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Cloud Shell

```
*****************************************************************************************
*****     Script              started             OCI_ComputeCapacityReport          *****
*****     Script              version             3.0.3                              *****
*****     Login               success             delegation_token                   *****
*****     Login               profile             /etc/oci/delegation_token          *****
*****     Tenancy             [redacted]          home region: FRA                   *****
*****     Region                                  eu-frankfurt-1                     *****
*****************************************************************************************

Do you have Administrator rights at the tenancy level? [Y]es, [N]o, [Q]uit: N

Enter a compartment OCID to which you have access or [Q]uit: [redacted]


Get all available shapes at: https://docs.oracle.com/en-us/iaas/Content/Compute/References/computeshapes.htm

BM.DenseIO.E4.128         BM.DenseIO.E5.128         BM.DenseIO1.36            BM.DenseIO2.52            BM.GPU.A10.4              BM.GPU.A100-v2.8
BM.GPU.B4.8               BM.GPU.H100.8             BM.GPU.L40S-NC.4          BM.GPU.L40S.4             BM.GPU2.2                 BM.GPU3.8
BM.GPU4.8                 BM.HPC.E5.144             BM.HPC2.36                BM.Optimized3.36          BM.Standard.A1.160        BM.Standard.B1.44
BM.Standard.E2.64         BM.Standard.E3.128        BM.Standard.E4.128        BM.Standard.E5.192        BM.Standard.E6.256        BM.Standard1.36
BM.Standard2.52           BM.Standard3.64           VM.DenseIO.E4.Flex        VM.DenseIO.E5.Flex        VM.DenseIO1.16            VM.DenseIO1.4
VM.DenseIO1.8             VM.DenseIO2.16            VM.DenseIO2.24            VM.DenseIO2.8             VM.GPU.A10.1              VM.GPU.A10.2
VM.GPU2.1                 VM.GPU3.1                 VM.GPU3.2                 VM.GPU3.4                 VM.Optimized3.Flex        VM.Standard.A1.Flex
VM.Standard.A2.Flex       VM.Standard.B1.1          VM.Standard.B1.16         VM.Standard.B1.2          VM.Standard.B1.4          VM.Standard.B1.8
VM.Standard.E2.1          VM.Standard.E2.1.Micro    VM.Standard.E2.2          VM.Standard.E2.4          VM.Standard.E2.8          VM.Standard.E3.Flex
VM.Standard.E4.Flex       VM.Standard.E5.Flex       VM.Standard.E6.Flex       VM.Standard1.1            VM.Standard1.16           VM.Standard1.2
VM.Standard1.4            VM.Standard1.8            VM.Standard2.1            VM.Standard2.16           VM.Standard2.2            VM.Standard2.24
VM.Standard2.4            VM.Standard2.8            VM.Standard3.Flex

Enter the name of a shape to discover its regional capacity or [Q]uit: VM.Standard.E6.Flex

Amount of OCPUs needed for shape VM.Standard.E6.Flex: 5

Specify the amount of memory (in GB) needed for shape VM.Standard.E6.Flex: 32

REGION            AVAILABILITY_DOMAIN        FAULT_DOMAIN       SHAPE                 OCPU    MEMORY    AVAILABILITY

eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-1   FAULT-DOMAIN-1     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-1   FAULT-DOMAIN-2     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-1   FAULT-DOMAIN-3     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-2   FAULT-DOMAIN-1     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-2   FAULT-DOMAIN-2     VM.Standard.E6.Flex   5       32        OUT_OF_HOST_CAPACITY
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-2   FAULT-DOMAIN-3     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-3   FAULT-DOMAIN-1     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-3   FAULT-DOMAIN-2     VM.Standard.E6.Flex   5       32        AVAILABLE
eu-frankfurt-1    fyxu:EU-FRANKFURT-1-AD-3   FAULT-DOMAIN-3     VM.Standard.E6.Flex   5       32        AVAILABLE

Enter another shape name, [P]rint shapes list or [Q]uit:
```

- **Back up (and define rollback)**
  - Take a **Boot Volume Backup** and **Block Volume Backups** (or application-level backups).
  - Define a rollback plan (common approach: keep the original instance stopped and unchanged until validation completes).
- **Stop the application gracefully**
  - Ensure applications and dependent services are shut down cleanly (drain connections, stop services, quiesce writes if needed).

3.2 In-place shape change (OCI Console)

- **Stop the instance**
  - Compute → Instances → select instance → **More actions** → **Stop**
  - Wait until lifecycle state shows **Stopped**.
- **Change the shape**
  - On the instance details page: **More actions** → **Change shape**
  - Select the new shape (for example **VM.Standard.E5.Flex** or **VM.Standard.E6.Flex**).
  - Configure:
    1. **OCPUs**
    2. **Memory (GB)** (Flex shapes allow adjusting within allowed ratios)

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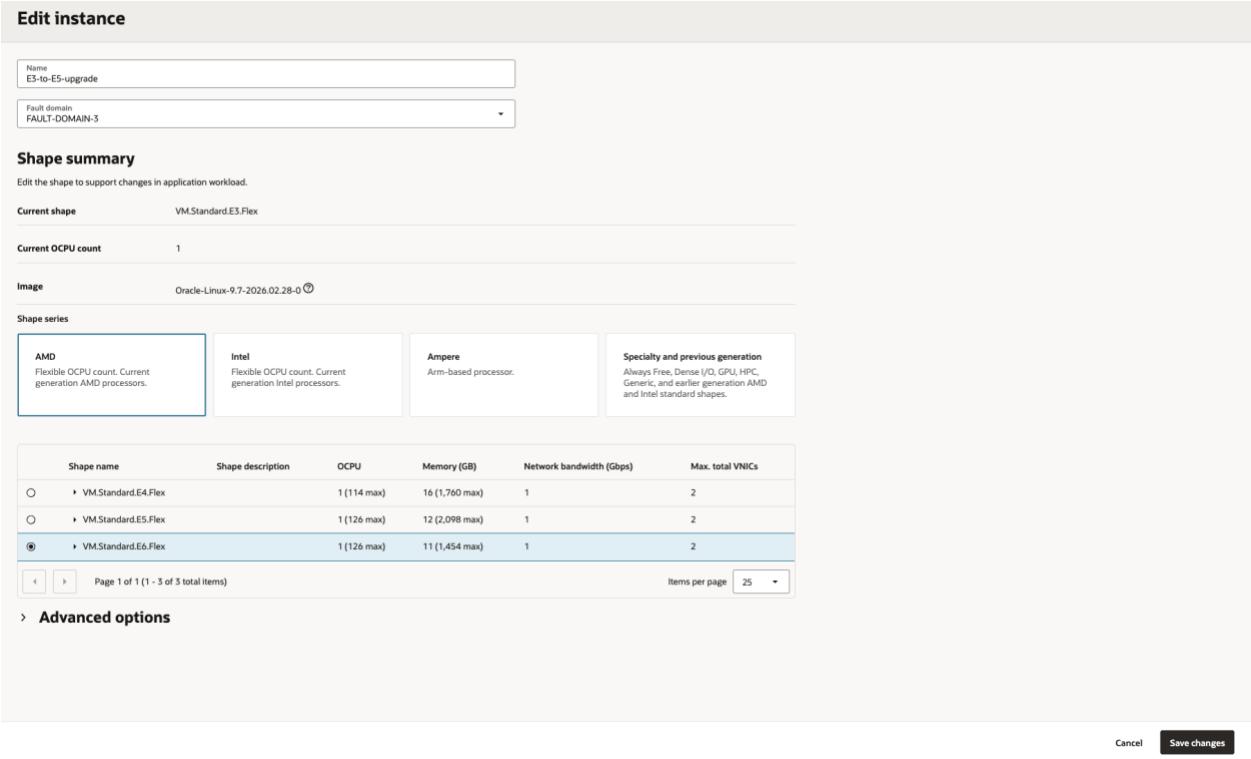

- **Confirm and apply**
  - Confirm the change and wait for OCI to complete the update.
- **Start the instance**
  - Click **Start**
  - Wait for lifecycle state to show **Running**.

3.3 Post-change validation

- **OS validation**: CPU/memory recognized correctly; system logs are clean; services start successfully.
- **Application validation**: run health checks/smoke tests; confirm performance and connectivity.
- **Network validation**:
  - Private IP should remain the same.
  - If using an **ephemeral public IP**, it may change after stop/start. Use a **Reserved Public IP** if you need a stable address.

Custom Image Method for Linux OS Instance (Replace Boot Volume)

This method applies when the instance is based on an image that was published **before** the targeted shapes were released. In such cases, customers performing an **in-place shape upgrade** may not see the **newer** shapes listed among the available options.

With this approach, customers can take advantage of the Replace Boot Volume feature, which allows Linux instances to replace their existing boot volume while moving or upgrading to a different instance shape. This makes it easier to migrate or upgrade instances without rebuilding the OS from scratch, helping reduce downtime and operational effort. It also enables customers to keep their existing system configuration and data on the boot volume while adopting a new shape that better matches performance or cost requirements.

In this example, we will demonstrate shape change from AMD E3 to AMD E6 by using Boot Volume replacement. In many use cases, it can be any shape to any shape based on requirements. As we underlined earlier, customers are advised to follow prerequisites carefully for successful execution and results.

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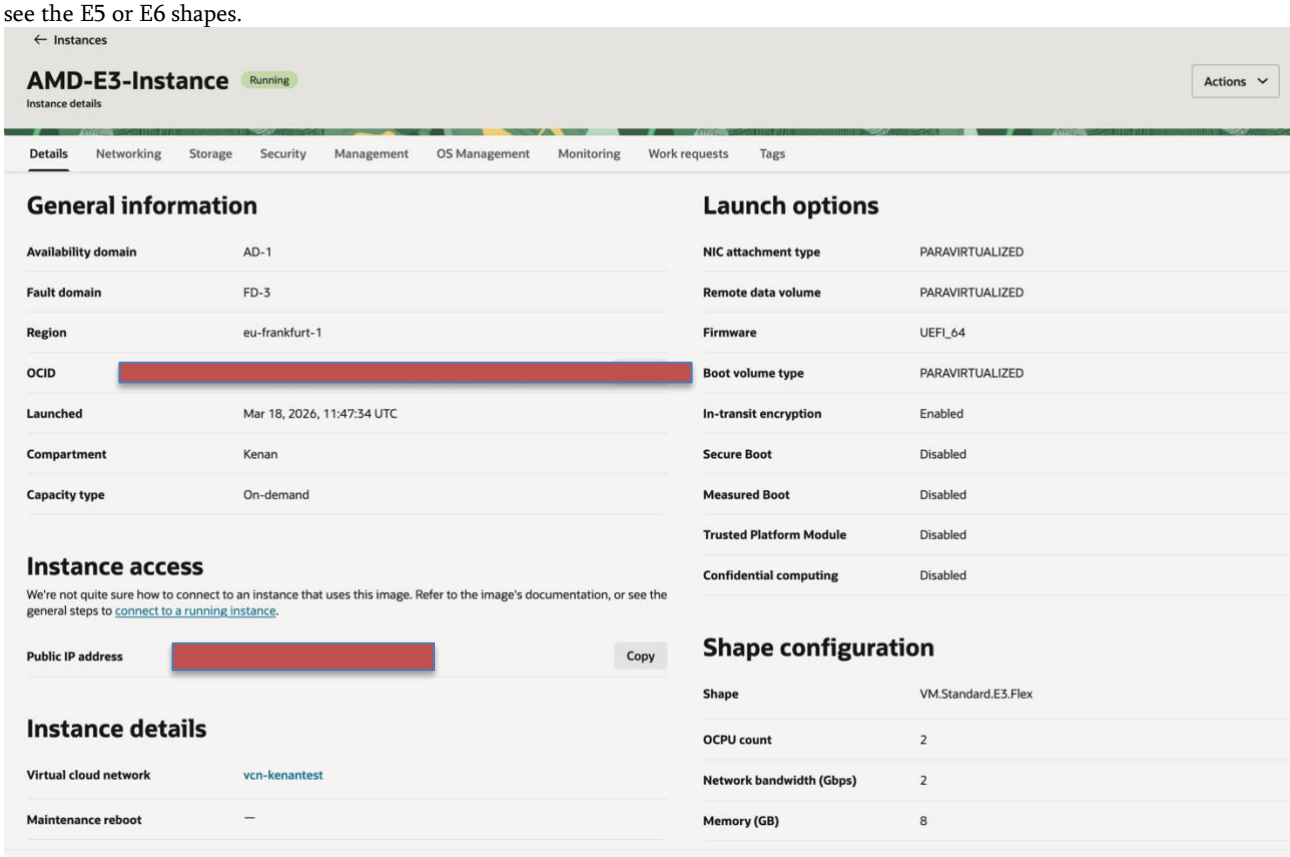
4.1 Pre-checks

- **Identify the current instance details**
  - OCI Console → **Compute** → **Instances** → select the instance. When you try to Edit and Upgrade Shape, you do not see the E5 or E6 shapes.



  - Current **shape**
  - **OCPUs / memory**
  - **Availability domain** (if applicable) and **fault domain**
  - **VNIC/subnet**, private IP, public IP (**ephemeral vs reserved**)
  - Attached **block volumes**, mount points, and any special boot configuration

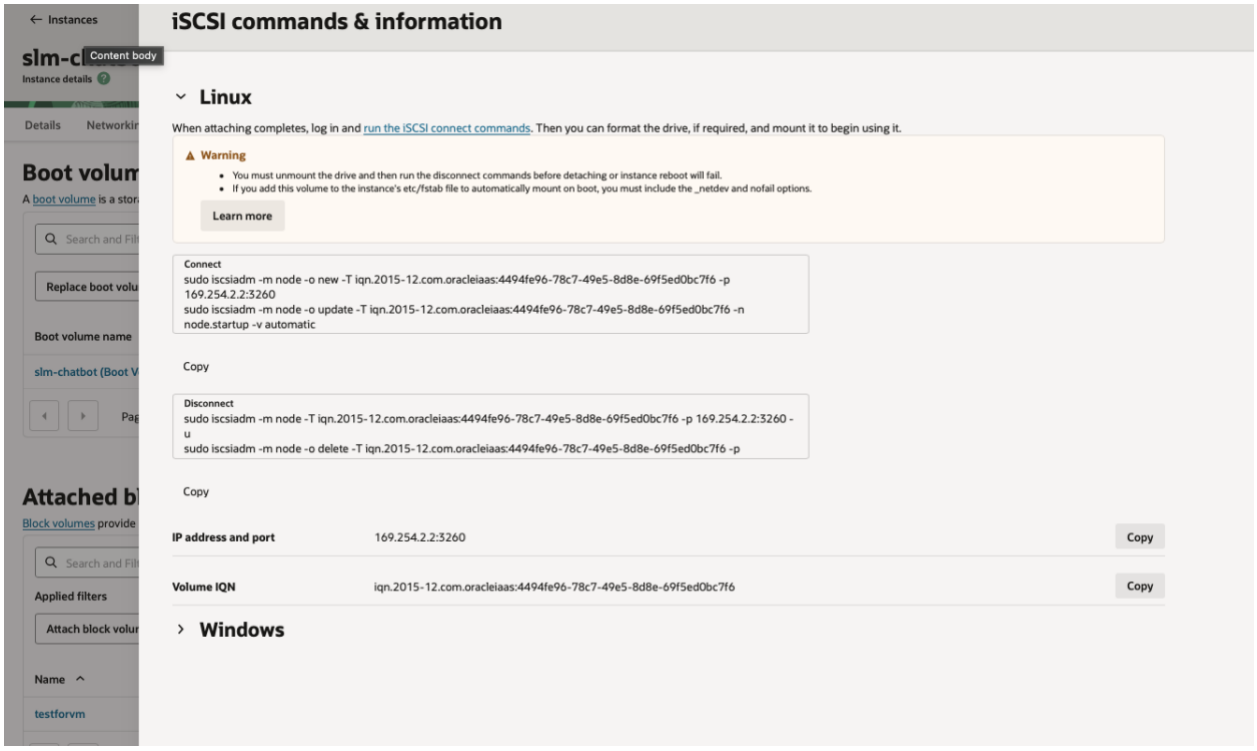
- **Confirm New Shape availability and limit**
  - Console → **Governance & Administration** → **Limits, Quotas and Usage**
  - Validate you have sufficient **OCPUs** for the target AMD E5 or E6 (**VM.Standard.E6.Flex**) in the correct **compartment/region/AD**.
  - Confirm the target shape is available in the instance's placement (some shapes are not available in every AD) using the OCI_ComputeCapacityReport.

Cloud Shell

```
*****************************************************************************************
*****     Script              started             OCI_ComputeCapacityReport          *****
*****     Script              version             3.0.3                              *****
*****     Login               success             delegation_token                   *****
*****     Login               profile             /etc/oci/delegation_token          *****
*****     Tenancy             [redacted]          home region: FRA                   *****
*****     Region              analyzed            eu-frankfurt-1                     *****
*****************************************************************************************

Do you have Administrator rights at the tenancy level? [Y]es, [N]o, [Q]uit: N

Enter a compartment OCID to which you have access or [Q]uit: [redacted]


Get all available shapes at: https://docs.oracle.com/en-us/iaas/Content/Compute/References/computeshapes.htm

BM.DenseIO.E4.128         BM.DenseIO.E5.128         BM.DenseIO1.36            BM.DenseIO2.52            BM.GPU.A10.4              BM.GPU.A100-v2.8
BM.GPU.B4.8               BM.GPU.H100.8             BM.GPU.L40S-NC.4          BM.GPU.L40S.4             BM.GPU2.2                 BM.GPU3.8
BM.GPU4.8                 BM.HPC.E5.144             BM.HPC2.36                BM.Optimized3.36          BM.Standard.A1.160        BM.Standard.B1.44
BM.Standard.E2.64         BM.Standard.E3.128        BM.Standard.E4.128        BM.Standard.E5.192        BM.Standard.E6.256        BM.Standard1.36
BM.Standard2.52           BM.Standard3.64           VM.DenseIO.E4.Flex        VM.DenseIO.E5.Flex        VM.DenseIO1.16            VM.DenseIO1.4
VM.DenseIO1.8             VM.DenseIO2.16            VM.DenseIO2.24            VM.DenseIO2.8             VM.GPU.A10.1              VM.GPU.A10.2
VM.GPU2.1                 VM.GPU3.1                 VM.GPU3.2                 VM.GPU3.4                 VM.Optimized3.Flex        VM.Standard.A1.Flex
VM.Standard.A2.Flex       VM.Standard.B1.1          VM.Standard.B1.16         VM.Standard.B1.2          VM.Standard.B1.4          VM.Standard.B1.8
VM.Standard.E2.1          VM.Standard.E2.1.Micro    VM.Standard.E2.2          VM.Standard.E2.4          VM.Standard.E2.8          VM.Standard.E3.Flex
VM.Standard.E4.Flex       VM.Standard.E5.Flex       VM.Standard.E6.Flex       VM.Standard1.1            VM.Standard1.16           VM.Standard1.2
VM.Standard1.4            VM.Standard1.8            VM.Standard2.1            VM.Standard2.16           VM.Standard2.2            VM.Standard2.24
VM.Standard2.4            VM.Standard2.8            VM.Standard3.Flex

Enter the name of a shape to discover its regional capacity or [Q]uit: VM.Standard.E6.Flex

Amount of OCPUs needed for shape VM.Standard.E6.Flex: 5

Specify the amount of memory (in GB) needed for shape VM.Standard.E6.Flex: 32

REGION              AVAILABILITY_DOMAIN         FAULT_DOMAIN        SHAPE                   OCPU     MEMORY    AVAILABILITY

eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-1    FAULT-DOMAIN-1      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-1    FAULT-DOMAIN-2      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-1    FAULT-DOMAIN-3      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-2    FAULT-DOMAIN-1      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-2    FAULT-DOMAIN-2      VM.Standard.E6.Flex     5        32        OUT_OF_HOST_CAPACITY
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-2    FAULT-DOMAIN-3      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-3    FAULT-DOMAIN-1      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-3    FAULT-DOMAIN-2      VM.Standard.E6.Flex     5        32        AVAILABLE
eu-frankfurt-1      fyxu:EU-FRANKFURT-1-AD-3    FAULT-DOMAIN-3      VM.Standard.E6.Flex     5        32        AVAILABLE

Enter another shape name, [P]rint shapes list or [Q]uit:
```

- **Back up (and define rollback)**
  - Take a **Boot Volume Backup** and **Block Volume Backups** (or application-level backups).
  - Define a rollback plan (common approach: keep the original instance stopped and unchanged until validation completes).
- **(Optional) Stop the application gracefully**
  - Ensure applications and dependent services are shut down cleanly (drain connections, stop services, quiesce writes if needed).
- **(OPTIONAL)Unmount and Detach additional Block Volumes**. We must safely unmount attached additional disks from old shape and we will attach it to new shape after successful upgrade. These steps are essential for safely disk unmounting process and avoid any data discrepancies.

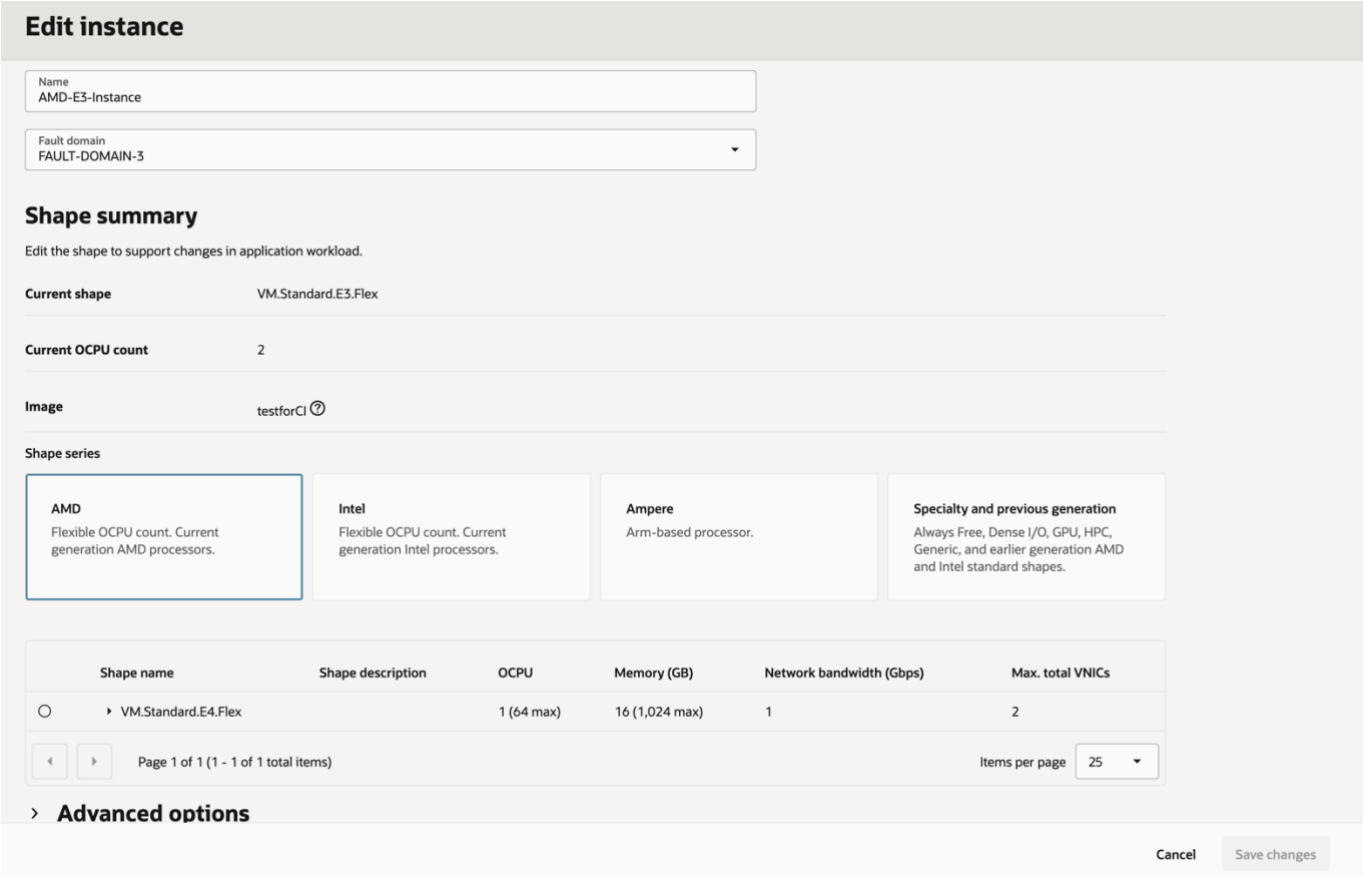
  - List attached disks with command **#lsblk** inside OS and identify disks (i.e. /dev/sdb, /dev/sdX)
  - Unmount the filesystem (#df -kh to list mount points) with command **# umount /mountpoint** → only attached Block Volumes. **!Not Boot Volume!**
  - Go to OCI Console
    - Select Instance Details and go to Storage tab
    - **Copy Disconnect ISCSI commands** from the page opens from 3 dots next to Block Volume and execute the commands in OS



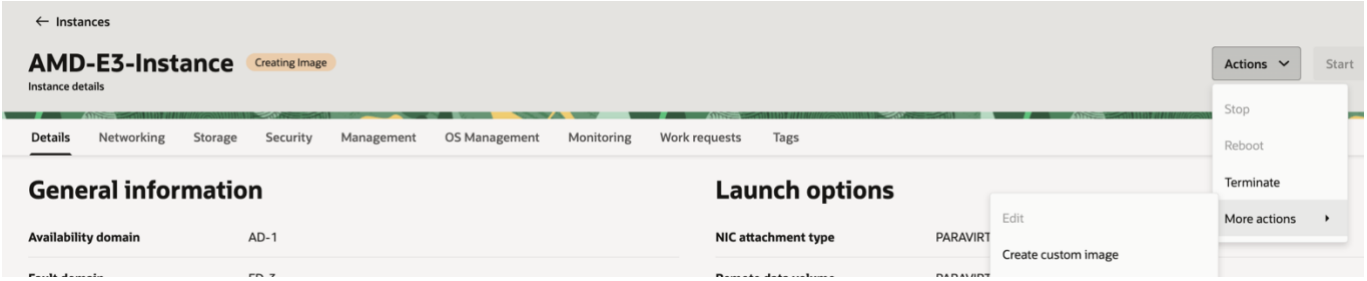
- After successful execution of commands in OS, disks will be removed from OS. We can check the disks by executing **#lsblk** again.
- Now it is safe to Detach Block Volume from the Instance. 3 Dots next to Block Volume Name and Detach.
- Comment out the entry of the disk from /etc/fstab for clean boot process.

4.2 Change Shape by Replacing Boot Volume

1. If we try to do In Place Shape Change, we will notice that targeted E5 or E6 shapes are not available in the list.



2. Go to Original Instance (The Instance you want to upgrade/change shape) → Actions → Create Custom Image and Provide a Name and Submit Create Custom Image. It will take some time to complete Create Custom Image process. **During the process, Instance will be in Shutdown state**. After completion, the instance will be reachable in Running state.



3. When Image is created, Go to **Compute → Custom Images** and note that Custom Image is available in the list.

4. Select **Custom Image → Actions → Edit Details** and Checkbox E5 and E6 shapes listed and Save Changes.

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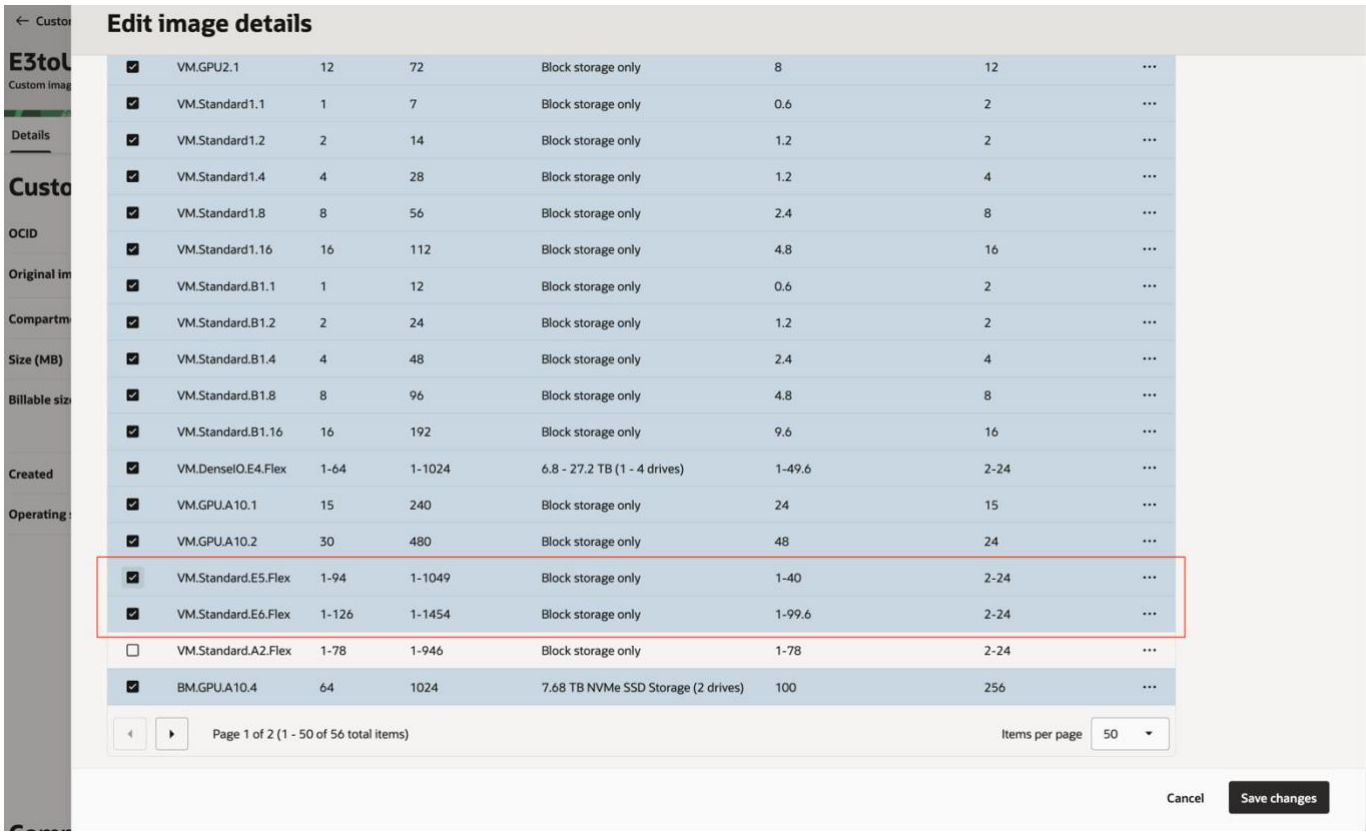

VM.GPU2.1	12	72	Block storage only	8	12	···
VM.Standard1.1	1	7	Block storage only	0.6	2	···
VM.Standard1.2	2	14	Block storage only	1.2	2	···
VM.Standard1.4	4	28	Block storage only	1.2	4	···
VM.Standard1.8	8	56	Block storage only	2.4	8	···
VM.Standard1.16	16	112	Block storage only	4.8	16	···
VM.Standard.B1.1	1	12	Block storage only	0.6	2	···
VM.Standard.B1.2	2	24	Block storage only	1.2	2	···
VM.Standard.B1.4	4	48	Block storage only	2.4	4	···
VM.Standard.B1.8	8	96	Block storage only	4.8	8	···
VM.Standard.B1.16	16	192	Block storage only	9.6	16	···
VM.DenseIO.E4.Flex	1-64	1-1024	6.8 - 27.2 TB (1 - 4 drives)	1-49.6	2-24	···
VM.GPU.A10.1	15	240	Block storage only	24	15	···
VM.GPU.A10.2	30	480	Block storage only	48	24	···
VM.Standard.E5.Flex	1-94	1-1049	Block storage only	1-40	2-24	···
VM.Standard.E6.Flex	1-126	1-1454	Block storage only	1-99.6	2-24	···
VM.Standard.A2.Flex	1-78	1-946	Block storage only	1-78	2-24	···
BM.GPU.A10.4	64	1024	7.68 TB NVMe SSD Storage (2 drives)	100	256	···

5. Go to Instances and Select Instance → Storage → Replace Boot Volume and Select Custom Image Name from dropdown list.
    a. Check Preserve Boot Volume (Delete later after successful operation)
    b. Select Replace by Image & Select Custom Image Name
    c. Submit by clicking Replace.
    d. It will take some time to replace the Boot Volume disks and **Instance will not be reachable during the process**.

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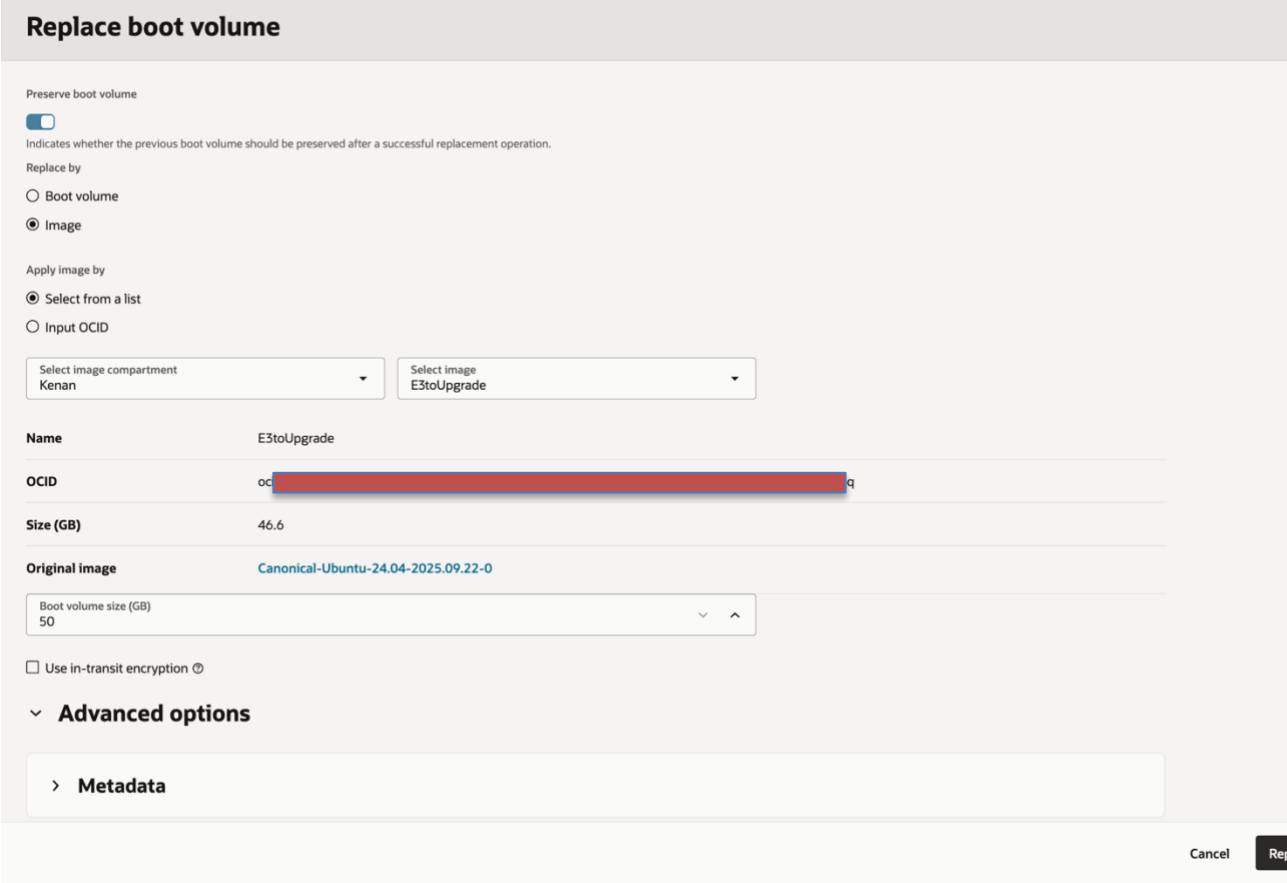

6. After Boot Volume replaced, the Instance will be in Running State and ready for Changing Current Shape to new Shape.
    a. Go to Edit Instance and Select New Shape (E6) and Modify OCPU and Memory as per requirement and Save Changes.
    b. Instance will be rebooted during the change and will be not reachable during that time.

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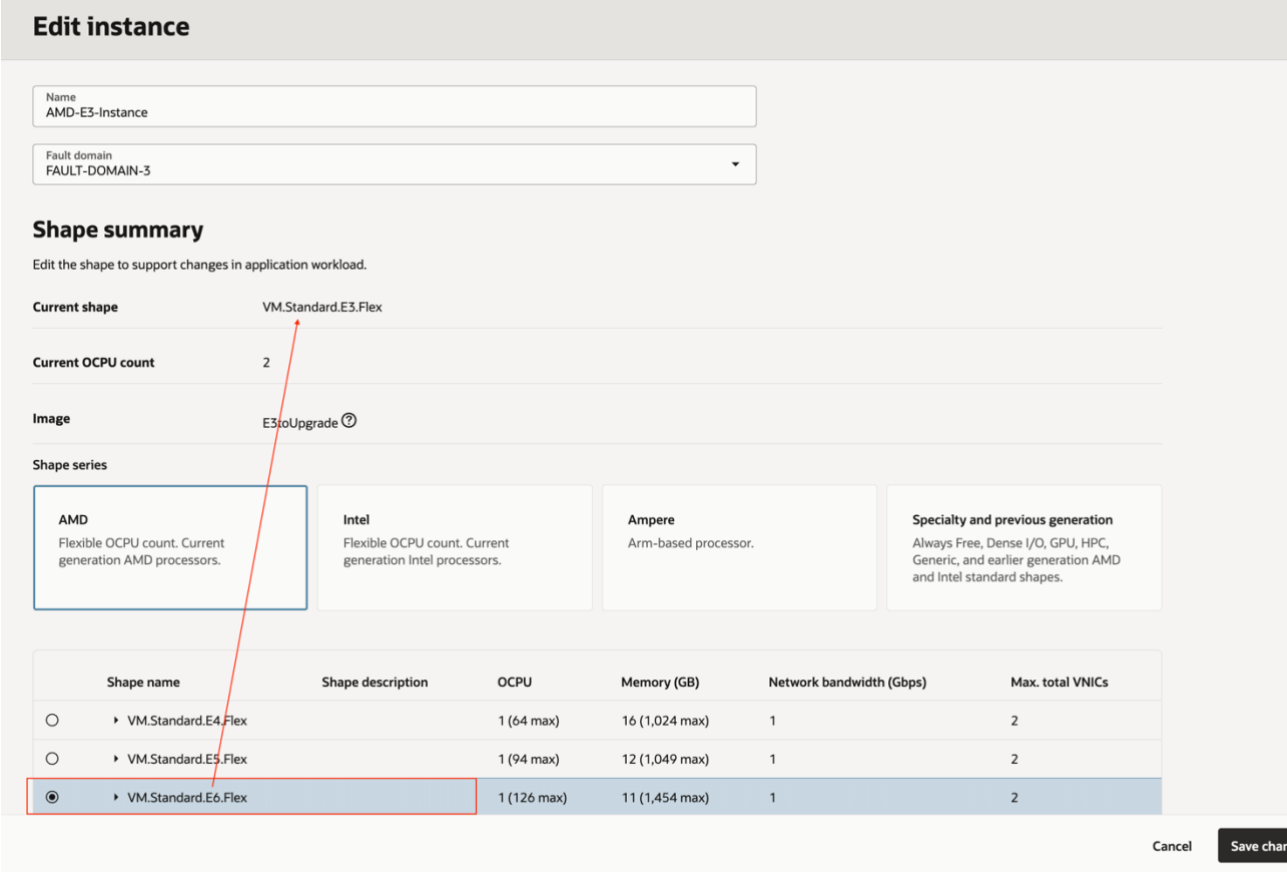
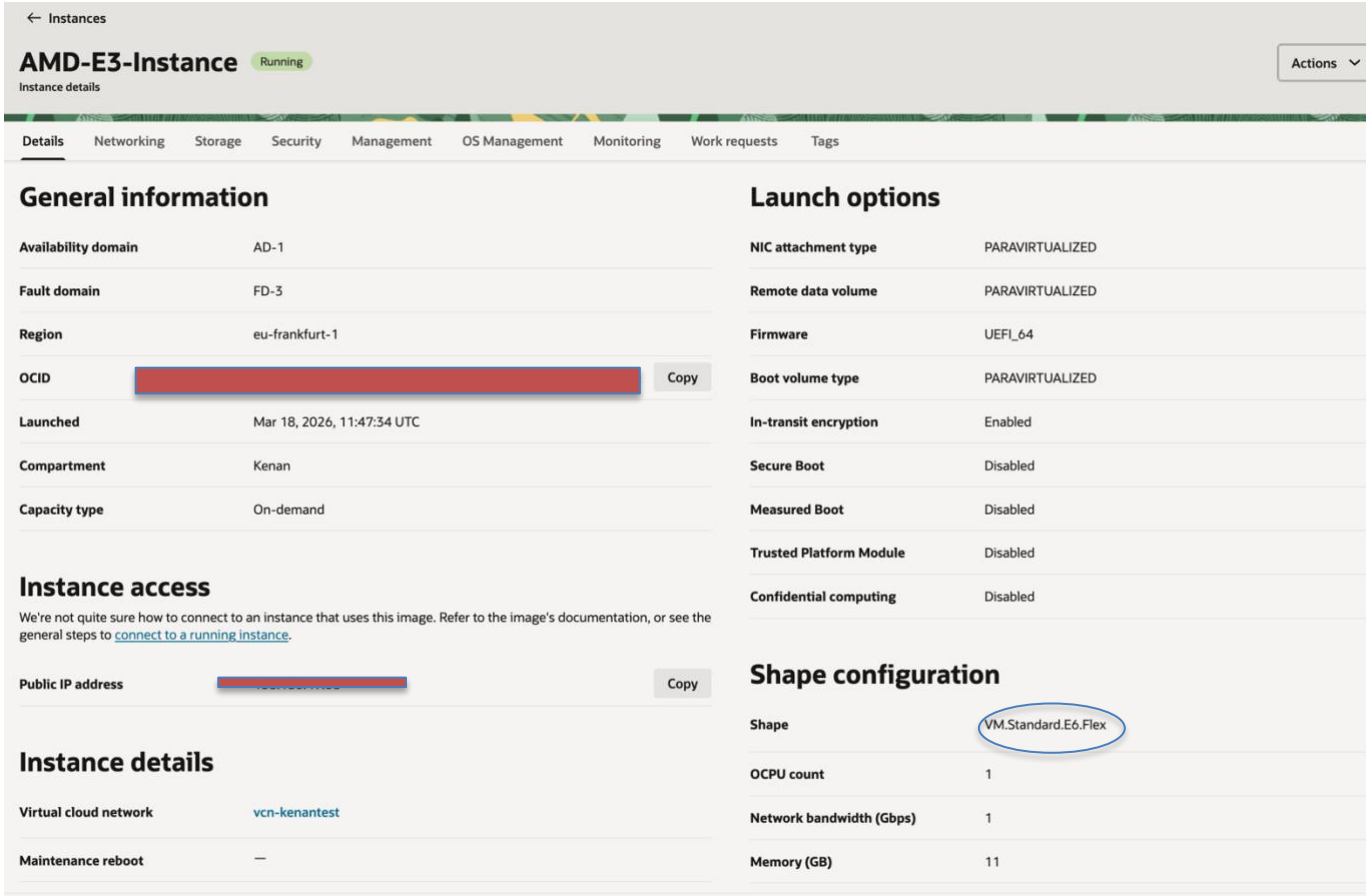


4.3 Post Activity Validation

- After instance is changed to new shape, you can connect via ssh with existing keys. As Boot Volume remains same, old keys will be persistent.



- **Private IP and MAC Address will be preserved in this method.**
- **OS validation**: CPU/memory recognized correctly; system logs are clean; services start successfully.
- **Application validation**: run health checks/smoke tests; confirm performance and connectivity.
- **(OPTIONAL)** To attach Additional Block Volumes, Go to OCI Console, Instance Details and Storage Tab, select Attach Block Volume and Select the required Volume with ISCSI attachment type.
    - After successful attachment, go to 3 dots next Block Volume name and copy ISCSI Connect Commands and paste to Linux Terminal.
    - As we have done during pre-checks, unmounting and arranging /etc/fstab entry, this time we need to mount disk with command such as # mount /dev/sdX and remove comment from /etc/fstab entry.
    - After mounting Block Volumes successfully, it is recommended to reboot Instance one time to see if it is booting up normally without issues.
- It is advised to check if OS and kernel updates are available.
- If the Instance is rebooting successfully, we can conclude the upgrade process of Instance.

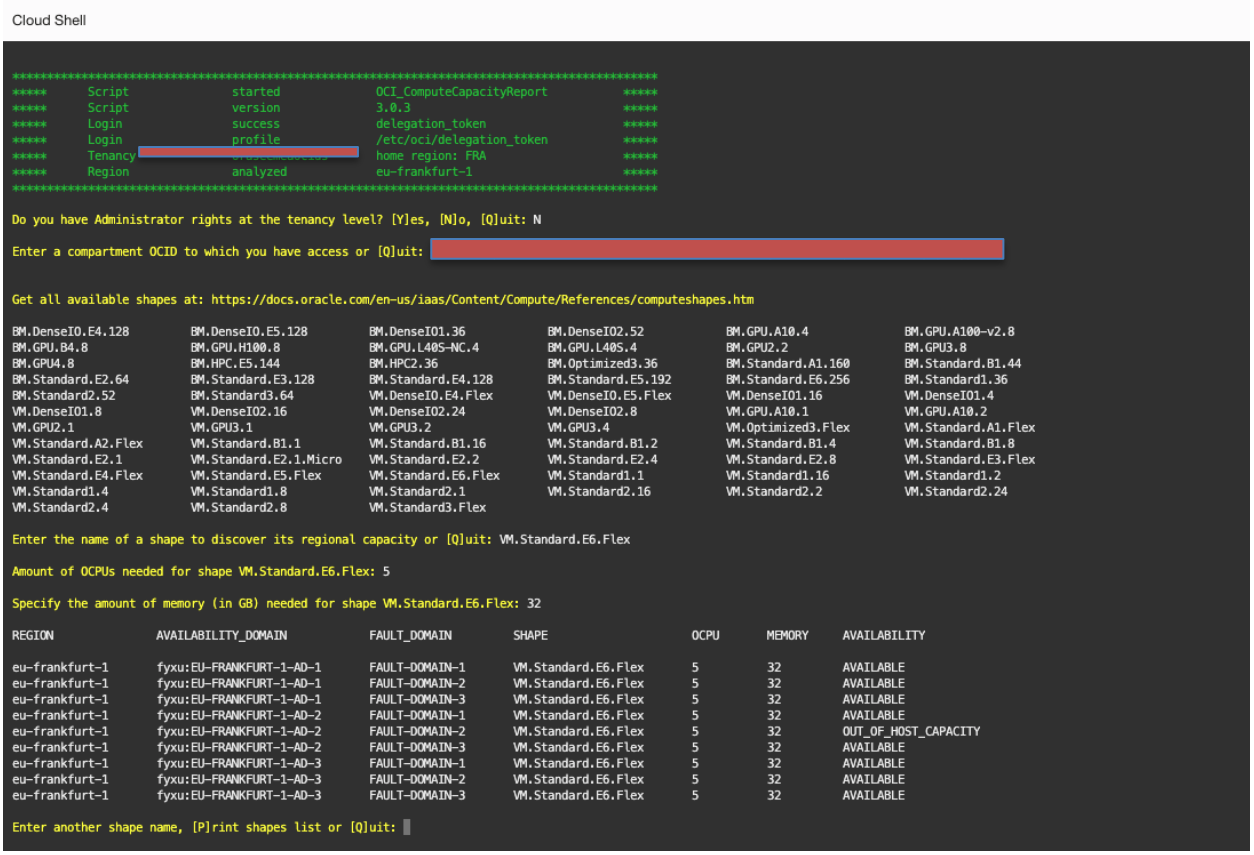
Custom Image Method for Windows OS Instances

This method applies when the instance is based on an image that was published **before** the targeted shapes were released. In such cases, customers performing an **in-place shape upgrade** may not see the **newer** shapes listed among the available options.

To address this, creating a **custom image** from the existing running instance and updating the image settings to include the **newer** shapes. In this example, we will use OCI E5/E6 shapes. Below is the step-by-step upgrade procedure.

5.1 Preparation

1. Identify the current instance details
   - OCI Console → Compute → Instances → select the instance
     Current shape
   - OCPUs / memory
   - Availability domain (if applicable) and fault domain
   - VNIC/subnet, private IP, public IP (ephemeral vs reserved)
   - Attached block volumes, mount points, and any special boot configuration
2. Confirm E5/E6 availability and limit
   - Console → Governance & Administration → Limits, Quotas and Usage
   - Validate you have sufficient OCPUs for the target E5/E6 shape in the correct compartment/region/AD.
   - Confirm the target shape is available in the instance's placement (some shapes are not available in every AD) using the OCI_ComputeCapacityReport.



3. **Reboot** the scoped Instance to see if it is booting up properly before the operation.
4. **Back up (and define rollback)**
   - Take a **Boot Volume Backup** and **Block Volume Backups** (or application-level backups).
   - Define a rollback plan (common approach: keep the original instance stopped and unchanged until validation completes).
5. **(OPTIONAL) Stop the application gracefully**
   - Ensure applications and dependent services are shut down cleanly (drain connections, stop services, quiesce writes if needed).

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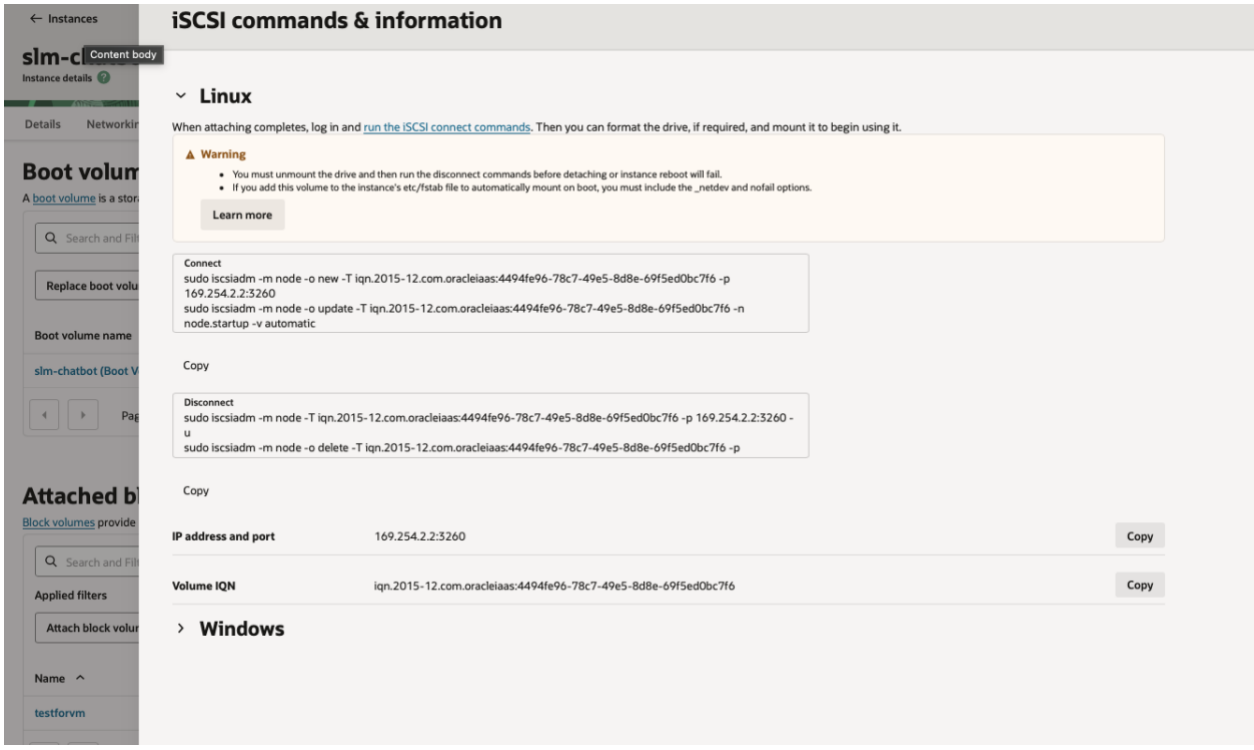
6. **(OPTIONAL) Unmount and Detach additional Block Volumes**. We must safely unmount attached additional disks from old shape and we will attach it to new shape after successful upgrade. These steps are essential for secure unmounting process and avoid any data discrepancies.

    - **For Linux Operating Systems** (you can refer this document more detail: Disconnecting from a Volume)

        - List attached disks with command **#lsblk** inside OS and identify disks (i.e. /dev/sdb, /dev/sdX)
        - Unmount the filesytem (#df -kh to list mount points) with command **# umount /mountpoint** → only attached Block Volumes. **!Not Boot Volume!**
        - Go to OCI Console
            - Select Instance Details and go to Storage tab
            - **Copy Disconnect ISCSI commands** from the page opens from 3 dots next to Block Volume and execute the commands in OS



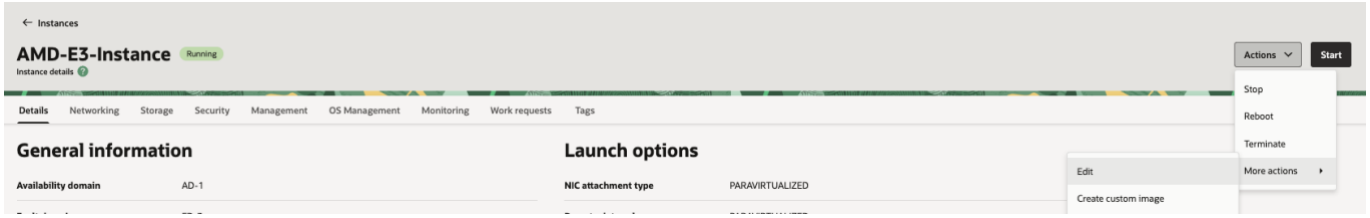
- After successful execution of commands in OS, disks will be removed from OS. We can check again by executing **#lsblk** again.
- Now it is safe to Detach Block Volume from Instance. 3 Dots next to Block Volume Name and Detach.

- **For Windows Operating Systems** , (you can refer this document more detail: Disconnecting from a Volume)

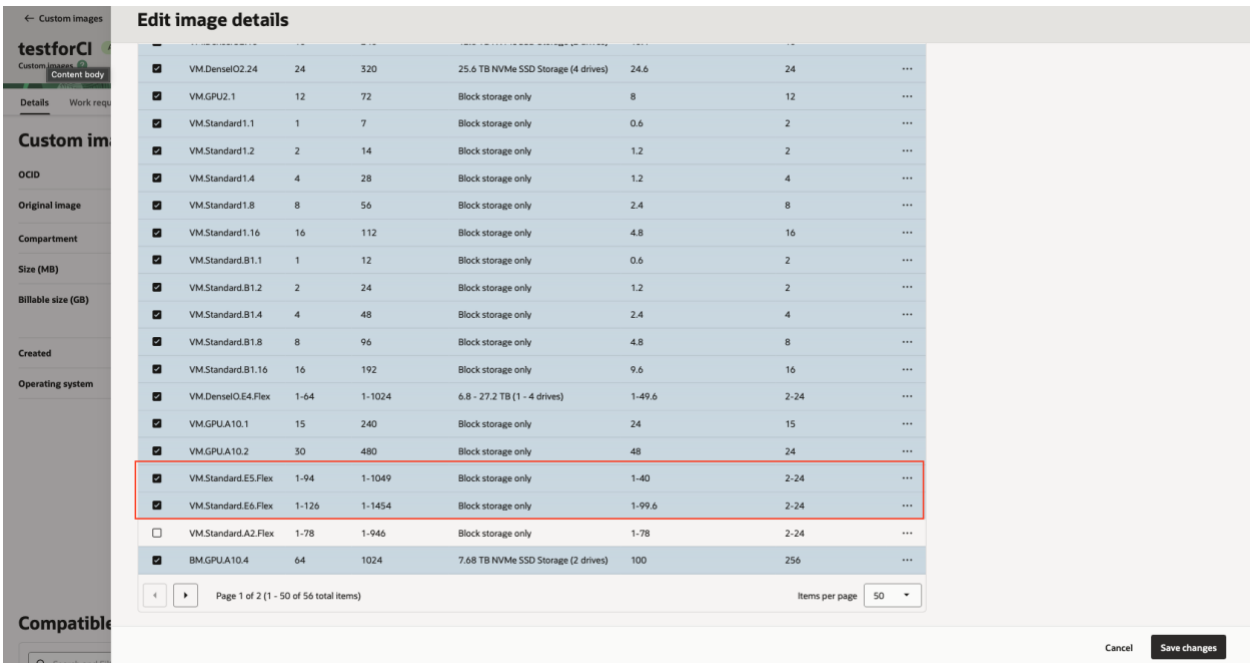
- Use a Remote Desktop client to log on to your Windows instance, and then open **Disk Management**.
- Right-select the volume you want to disconnect, and then select **Offline**. (not Boot Volume)
- Open **iSCSI Initiator**, select the target, and then select **Disconnect**.
- Confirm the session termination. The status should show as **Inactive**.
- In **iSCSI Initiator**, select the **Favorite Targets** tab, select the target you are disconnecting, and then select **Remove**.
- Select the **Volumes and Devices** tab, select the volume from the **Volume List**, and then select **Remove**.
- You can now detach the volume without the risk of losing data.

5.2 Custom Image and New Instance Creation

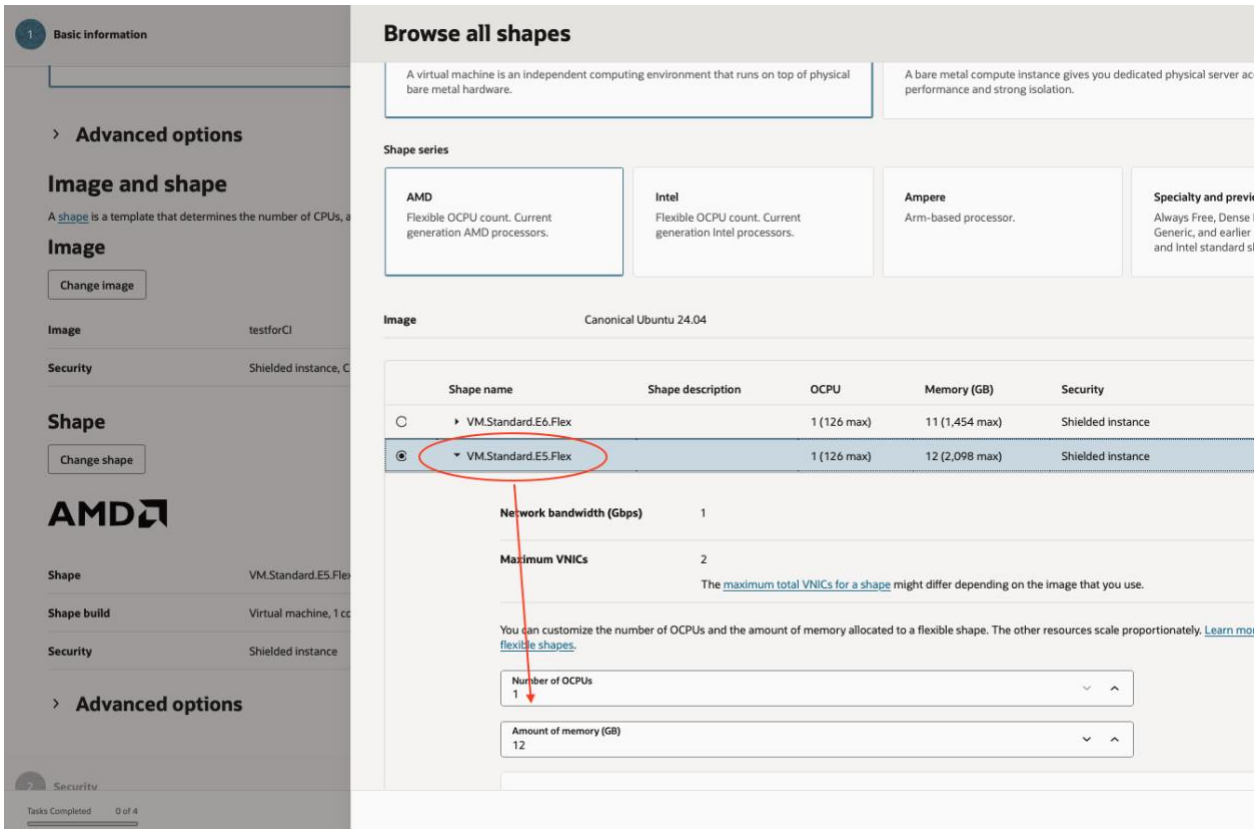
- **Shutdown** Instance
- Select **Create Custom Image** Under Actions



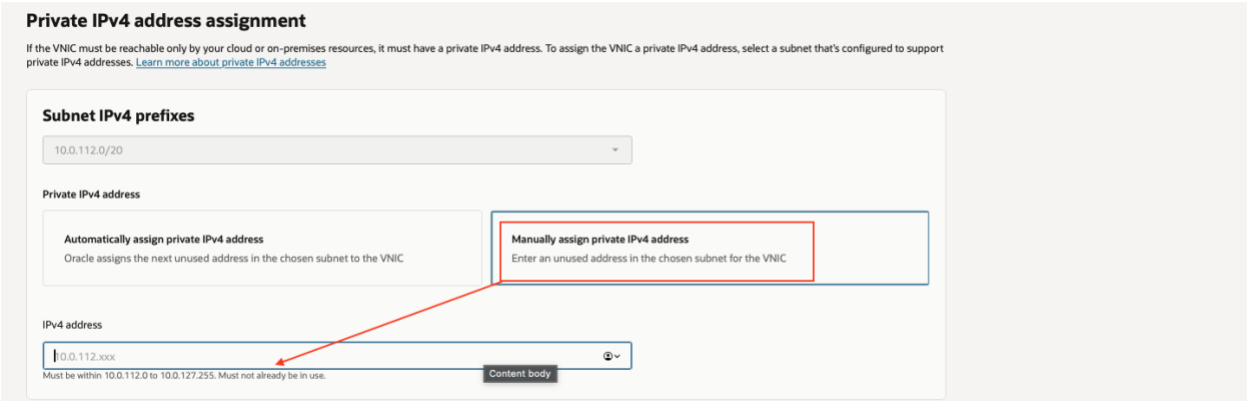
- Provide a Name for Custom Image and Submit.
- It will take to finish 10-20 mins depending on volume size.
- After successful work request, new Custom Image will be available under Custom Images.
- Go to **Custom Image details** → **Edit Details** → **Add E5/E6 shapes** and **Save**.



- Create a new Instance from this Custom image. This time VM.Standart.E5.Flex and VM.Standart.E6.Flex shapes will be available under Shape selection menu. OCPU and Memory can be adjusted too.



- The remaining steps are similar as a regular Instance Creation process.
- **!Important! If you want to retain the PRIVATE IP address of existing Instance, you have to Terminate existing (Instance with Old Shape) to free up IP Address. Therefore, you can manually assign the same IP address as old Instance.**



- For Reserved Public IP, we can Assigned Reserved Public IP from existing Reserved IPs.
- Existing passwords and user credentials will work for Windows OS Instances because the Instance Boot volume has keys will be preserved.
- We can attach the Additional Block Volumes after successful Instance creation with Healthy State.
- After selecting, Image, Shape, Network, IP Address and Boot volume. We can create the new Instance.

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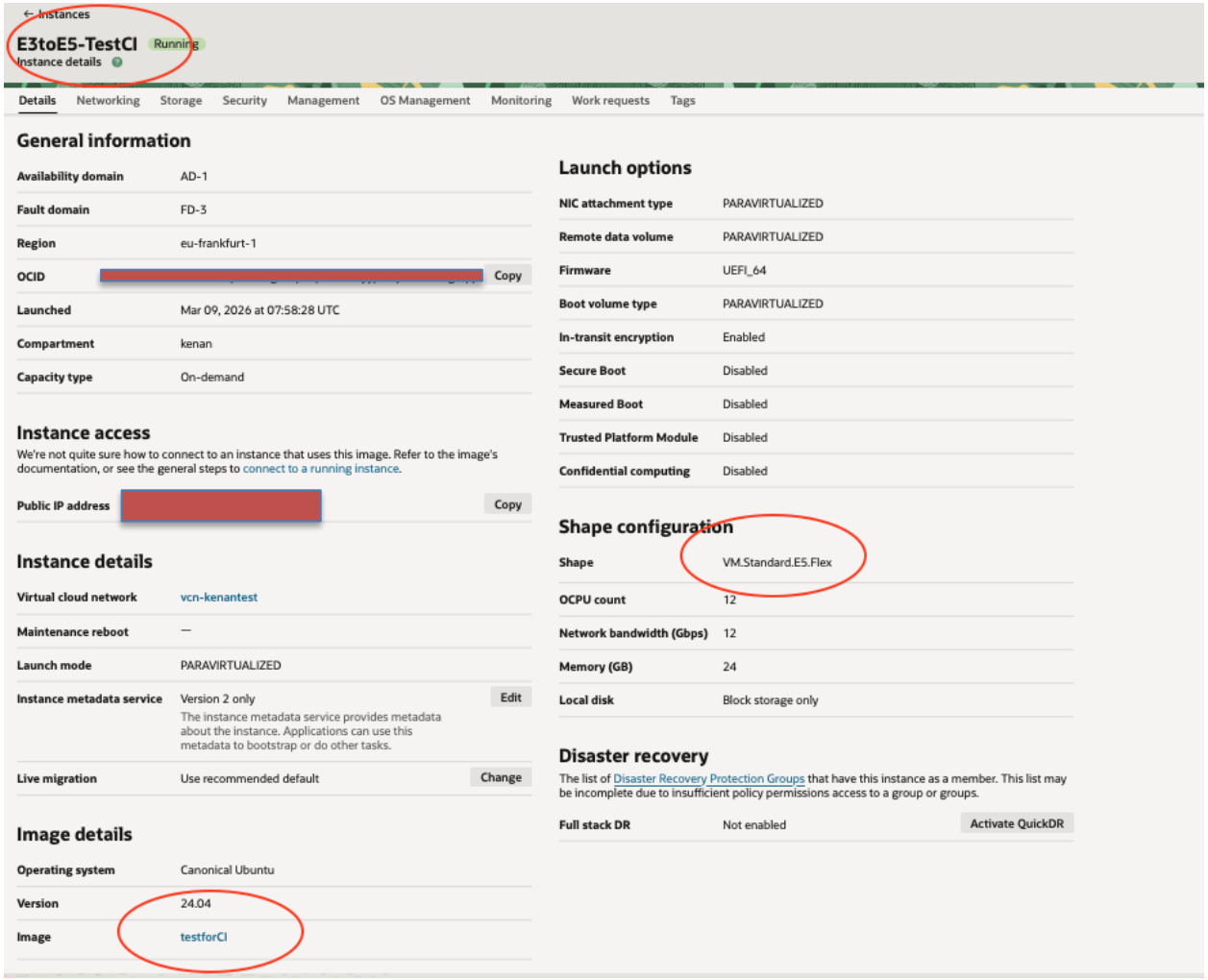
5.3 Post Instance Creation Validation

- After New instance with E5/E6 shape is created, we can connect RDP with same password for Windows OS. Since there is no change in boot volume, ssh keys and Windows OS user credentials will remain same.
- (Optional) To attach Additional Block Volumes for Windows,
  - Go to OCI Console, Instance Details and Storage Tab, select Attach Block Volume and Select the required Volume with ISCSI attachment type.
  - After successful attachment, go to 3 dots next BV name and copy Connect Commands and paste into Powershell Terminal in OS.
- After mounting Block Volumes successfully, it is recommended to reboot Instance one time to see if it is booting up normally without issues.
- It is advised to check if OS and kernel updates are available.
- If the Instance is rebooting successfully, we can conclude the upgrade process of Instance.



Upgrading Existing Shapes to New Shapes with Acceleron

OCI has recently introduced new Acceleron (Ax) Compute shapes. Moving from existing shapes (for example E4/E5, X9, or earlier generations) to Ax shapes requires a defined upgrade approach.
There are two supported methods:

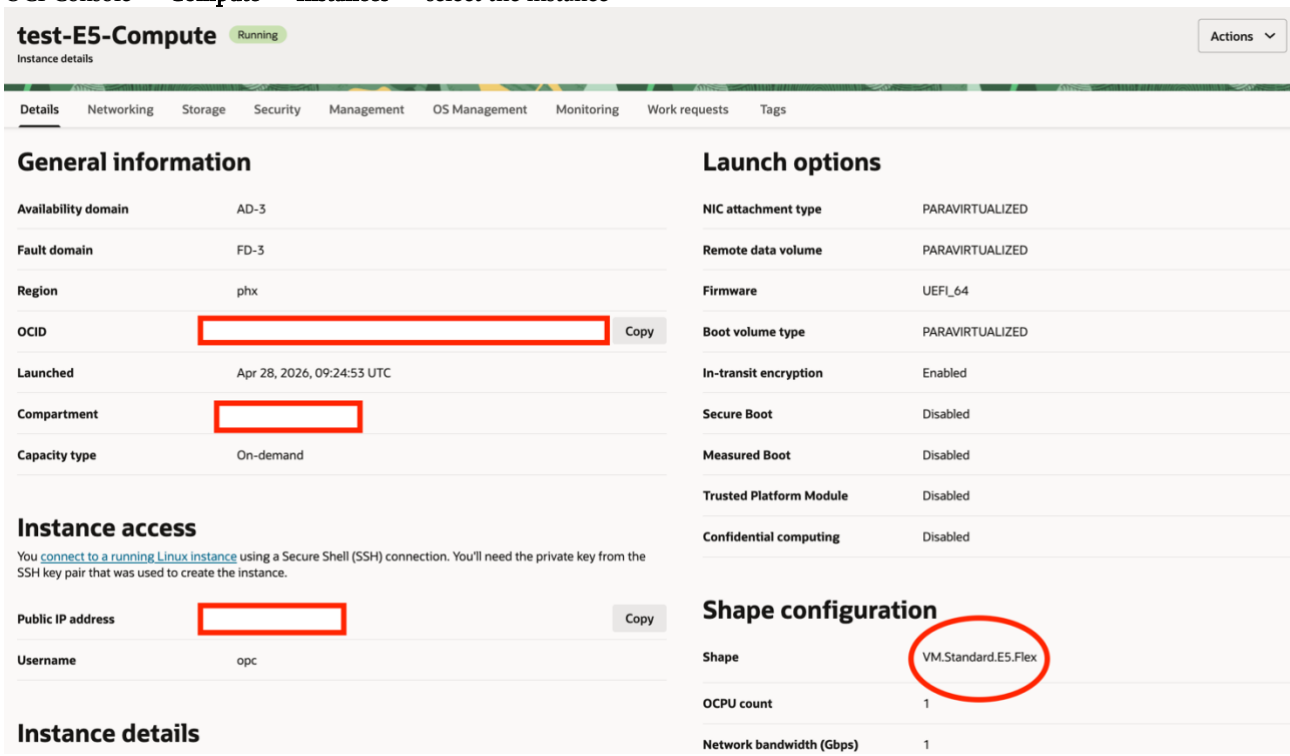
- In-place shape upgrade: Change the shape of an existing instance to a compatible Acceleron (Ax) shape. The instance is stopped, updated, and restarted with the new shape while keeping the same boot volume and instance OCID.
- Boot volume replacement (for Linux Instances): Launch a new instance using an existing boot volume on an Acceleron (Ax) shape. This method is used when in-place upgrade is not supported or when additional control over the migration is needed.

This topic describes both methods and when to use each.

In Place Shape Upgrade Method

6.1 Pre-checks for In Place Upgrade

- **Identify the current instance details**
  - OCI Console → **Compute** → **Instances** → select the instance



  - Current **shape**
  - **OCPUs / memory**
  - **Availability domain** (if applicable) and **fault domain**
  - **VNIC/subnet**, private IP, public IP (**ephemeral vs reserved**)
  - Attached **block volumes**, mount points, and any special boot configuration
- **Confirm New Shape availability and limit**
  - Console → **Governance & Administration** → **Limits, Quotas and Usage**
  - Validate you have sufficient **OCPUs** for the target E6 Ax Flex shape in the correct **compartment/region/AD**.
  - Confirm the target shape is available in the instance's placement (some shapes are not available in every AD) using the OCI_ComputeCapacityReport.

```
Enter another shape name, [P]rint shapes list or [Q]uit: VM.Standard.E6.Ax.Flex

Amount of OCPUs needed for shape VM.Standard.E6.Ax.Flex: 1

Specify the amount of memory (in GB) needed for shape VM.Standard.E6.Ax.Flex: 1

REGION          AVAILABILITY_DOMAIN     FAULT_DOMAIN     SHAPE                OCPU    MEMORY    AVAILABILITY

us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         OUT_OF_HOST_CAPACITY
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         AVAILABLE
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         AVAILABLE

Enter another shape name, [P]rint shapes list or [Q]uit:
```

- **Back up (and define rollback)**
  - Take a **Boot Volume Backup** and **Block Volume Backups** (or application-level backups).
  - Define a rollback plan (common approach: keep the original instance stopped and unchanged until validation completes).
- **Stop the application gracefully**
  - Ensure applications and dependent services are shut down cleanly (drain connections, stop services, quiesce writes if needed).

6.2 In-place shape change (OCI Console)

- **Stop the instance**
  - Compute → Instances → select instance → **More actions** → **Stop**
  - Wait until lifecycle state shows **Stopped**.
- **Change the shape**
  - On the instance details page: **More actions** → **Change shape**
  - Select the new shape (for example **VM.Standard.E6.Ax.Flex**).
  - Configure:
    1. **OCPUs**
    2. **Memory (GB)** (Flex shapes allow adjusting within allowed ratios)

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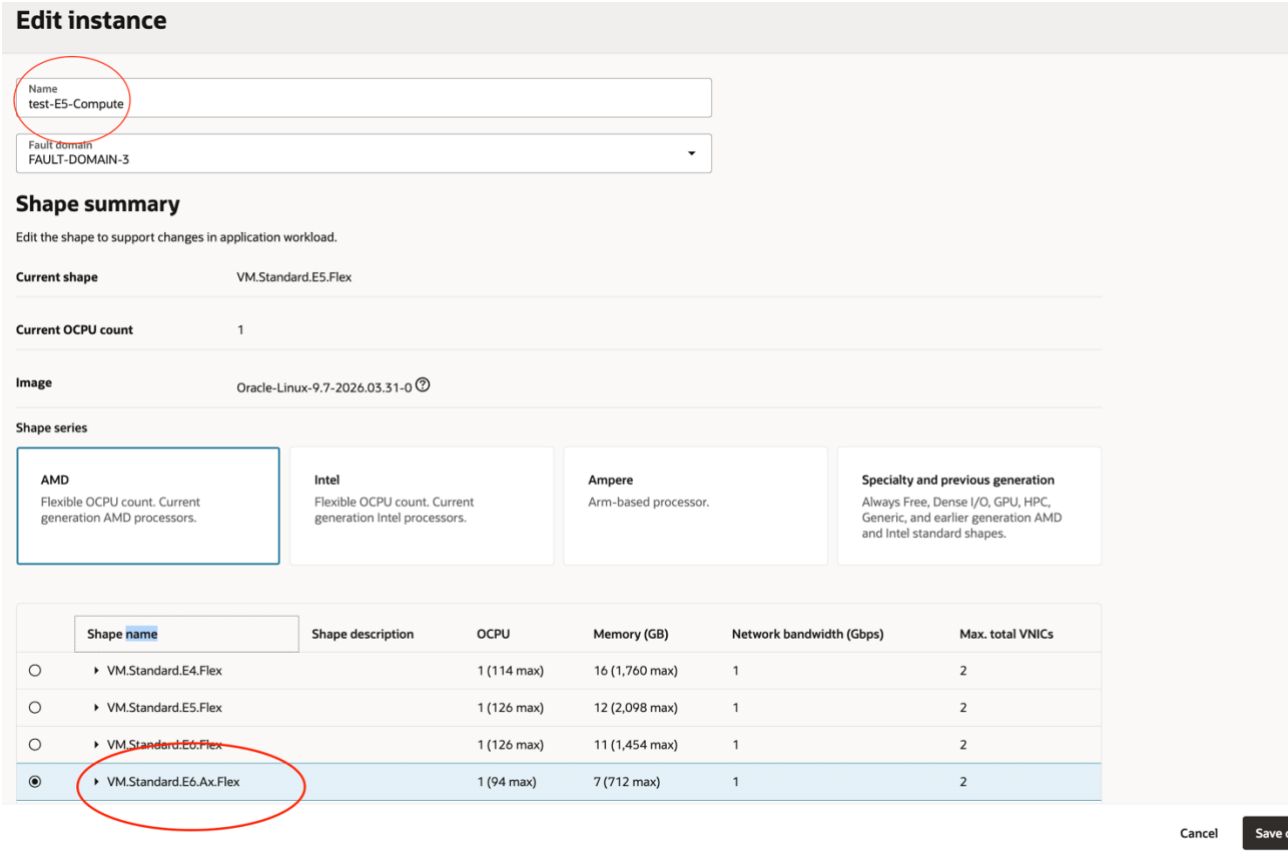

- **Confirm and apply**
  - Confirm the change and wait for OCI to complete the update.
- **Start the instance**
  - Click **Start**
  - Wait for lifecycle state to show **Running**.

6.3 Post-change validation

- **OS validation**: CPU/memory recognized correctly; system logs are clean; services start successfully.
- **Application validation**: run health checks/smoke tests; confirm performance and connectivity.
- **Network validation**:
  - Private IP should remain the same.
  - If using an **ephemeral public IP**, it may change after stop/start. Use a **Reserved Public IP** if you need a stable address.

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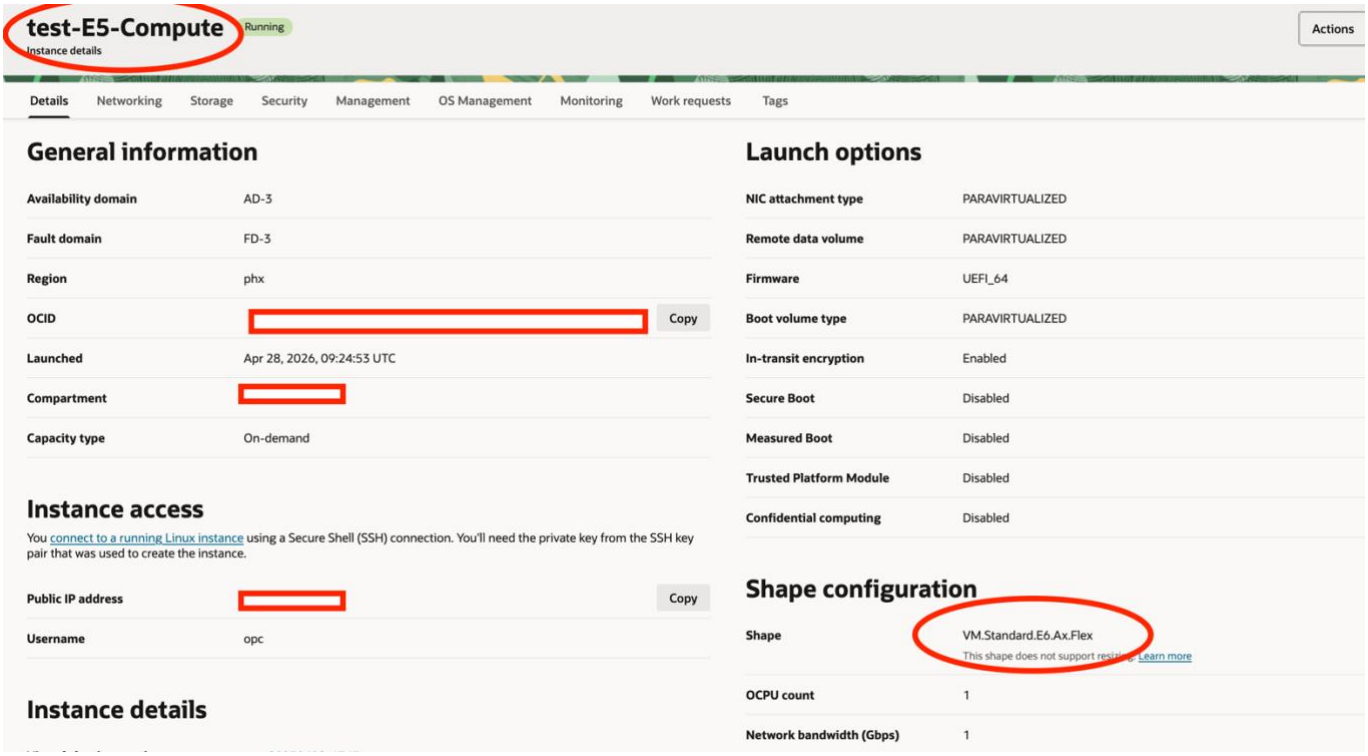
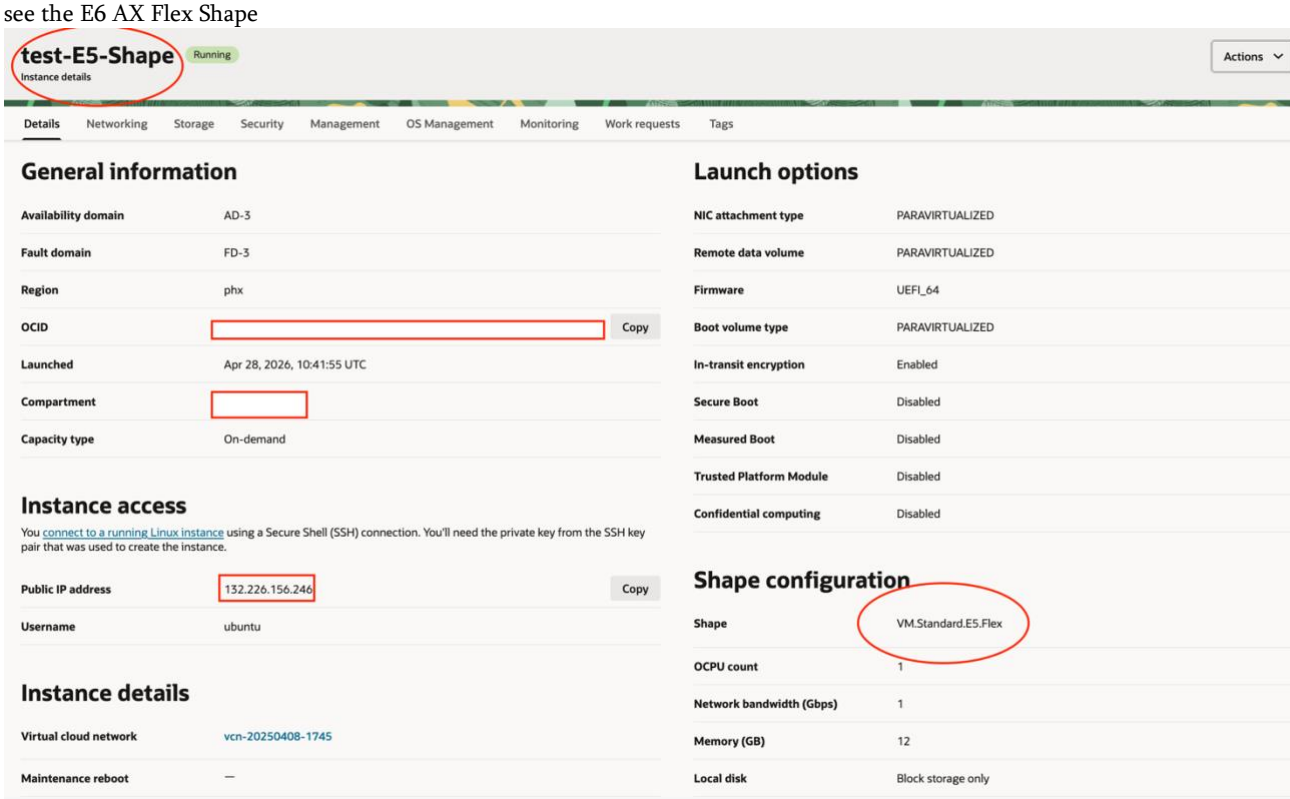


Boot volume replacement (for Linux Instances)

6.4 Pre-checks

- **Identify the current instance details**

- OCI Console → **Compute** → **Instances** → select the instance. When you try to Edit and Upgrade Shape, you do not see the E6 AX Flex Shape



- Current **shape**
- **OCPUs / memory**
- **Availability domain** (if applicable) and **fault domain**
- **VNIC/subnet**, private IP, public IP (**ephemeral vs reserved**)
- Attached **block volumes**, mount points, and any special boot configuration

- **Confirm New Shape availability and limit**

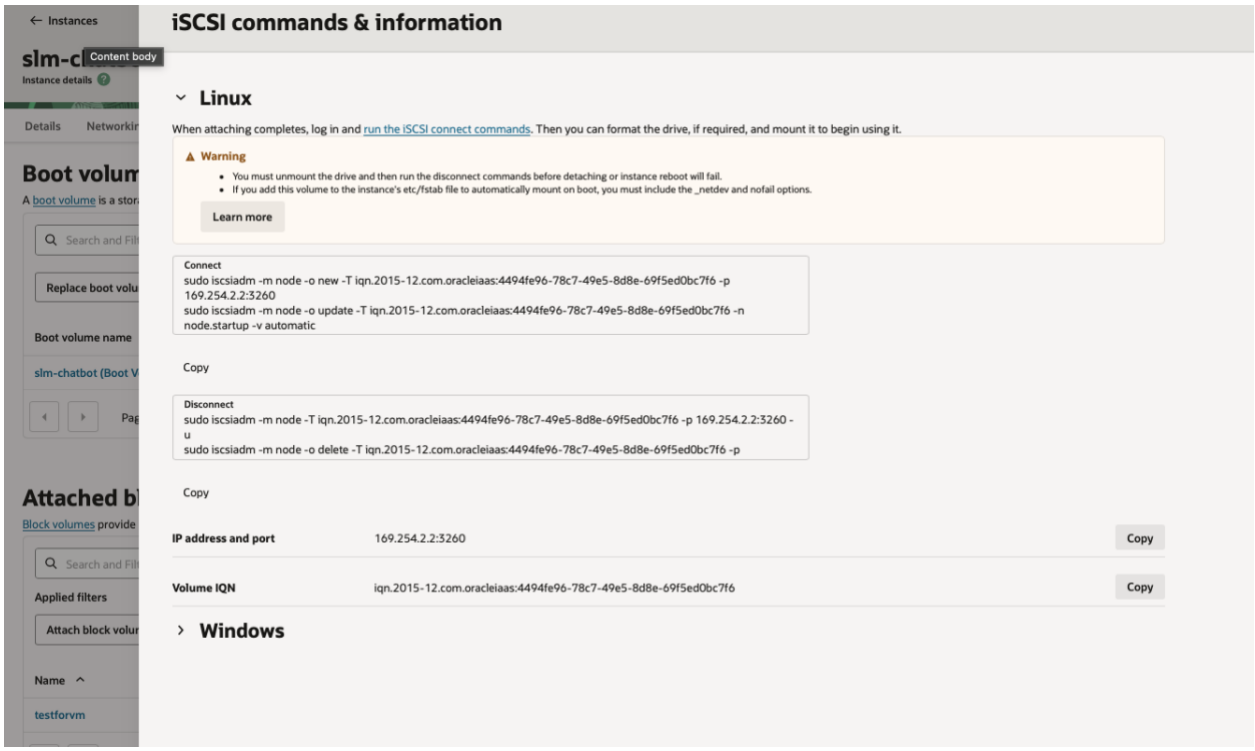
  - Console → **Governance & Administration** → **Limits, Quotas and Usage**
  - Validate you have sufficient **OCPUs** for the target AMD E5 or E6 (**VM.Standard.E6.Flex**) in the correct **compartment/region/AD**.
  - Confirm the target shape is available in the instance's placement (some shapes are not available in every AD) using the OCI_ComputeCapacityReport.
  -

```
Enter another shape name, [P]rint shapes list or [Q]uit: VM.Standard.E6.Ax.Flex

Amount of OCPUs needed for shape VM.Standard.E6.Ax.Flex: 1

Specify the amount of memory (in GB) needed for shape VM.Standard.E6.Ax.Flex: 1

REGION          AVAILABILITY_DOMAIN     FAULT_DOMAIN     SHAPE                OCPU    MEMORY    AVAILABILITY

us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-1           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-2           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         HARDWARE_NOT_SUPPORTED
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-1   VM.Standard.E6.Ax.Flex  1       1         OUT_OF_HOST_CAPACITY
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-2   VM.Standard.E6.Ax.Flex  1       1         AVAILABLE
us-phoenix-1    fyxu:PHX-AD-3           FAULT-DOMAIN-3   VM.Standard.E6.Ax.Flex  1       1         AVAILABLE

Enter another shape name, [P]rint shapes list or [Q]uit:
```

- **Back up (and define rollback)**
  - Take a **Boot Volume Backup** and **Block Volume Backups** (or application-level backups).

  - Define a rollback plan (common approach: keep the original instance stopped and unchanged until validation completes).
- **(Optional) Stop the application gracefully**
  - Ensure applications and dependent services are shut down cleanly (drain connections, stop services, quiesce writes if needed).
- **(OPTIONAL)Unmount and Detach additional Block Volumes**. We must safely unmount attached additional disks from old shape and we will attach it to new shape after successful upgrade. These steps are essential for safely disk unmounting process and avoid any data discrepancies.

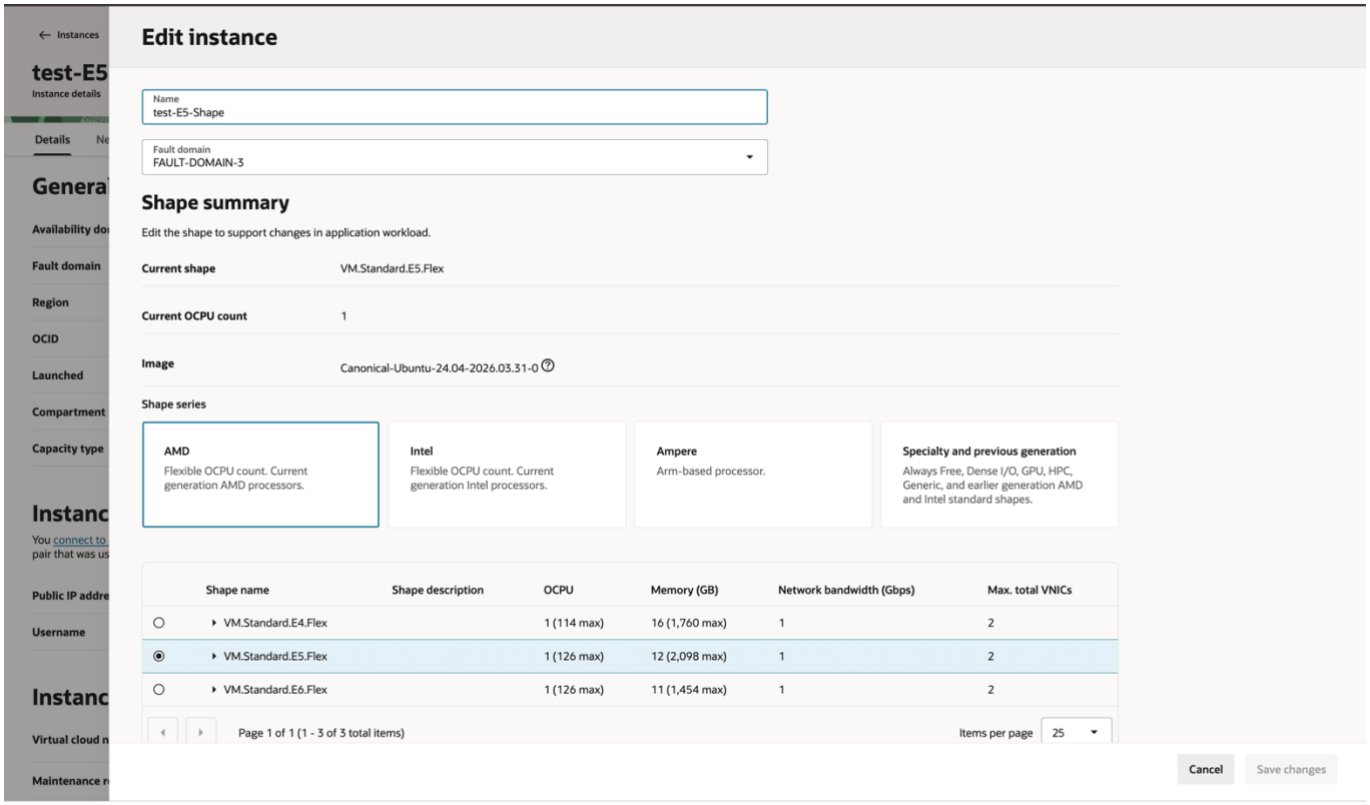
    - List attached disks with command **#lsblk** inside OS and identify disks (i.e. /dev/sdb, /dev/sdX)
    - Unmount the filesystem (#df -kh to list mount points) with command **# umount /mountpoint** → only attached Block Volumes. **!Not Boot Volume!**
    - Go to OCI Console
      - Select Instance Details and go to Storage tab
      - **Copy Disconnect ISCSI commands** from the page opens from 3 dots next to Block Volume and execute the commands in OS



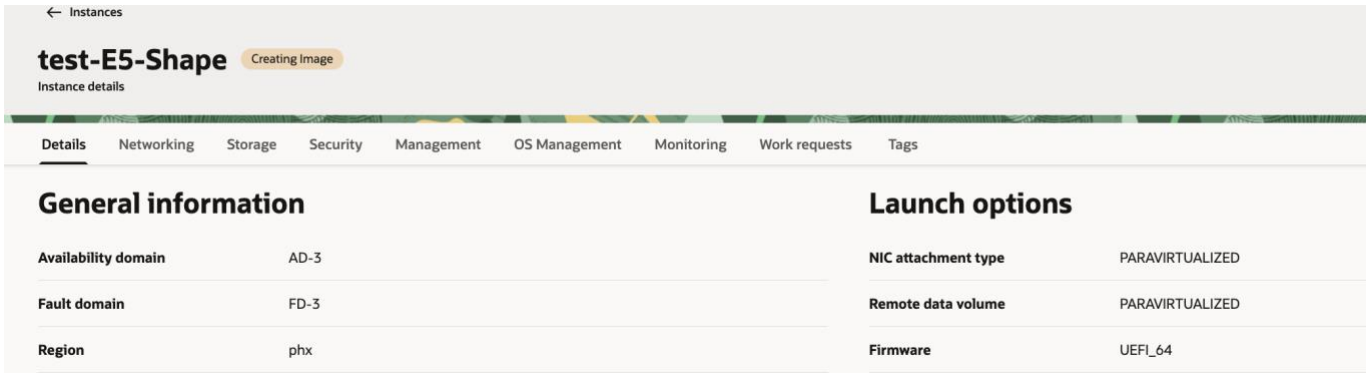
- After successful execution of commands in OS, disks will be removed from OS. We can check the disks by executing **#lsblk** again.
- Now it is safe to Detach Block Volume from the Instance. 3 Dots next to Block Volume Name and Detach.
- Comment out the entry of the disk from /etc/fstab for clean boot process.

6.5 Change Shape by Replacing Boot Volume

7. If we try to do In Place Shape Change, we will notice that targeted E6 Ax Flex shape is not available in the list.

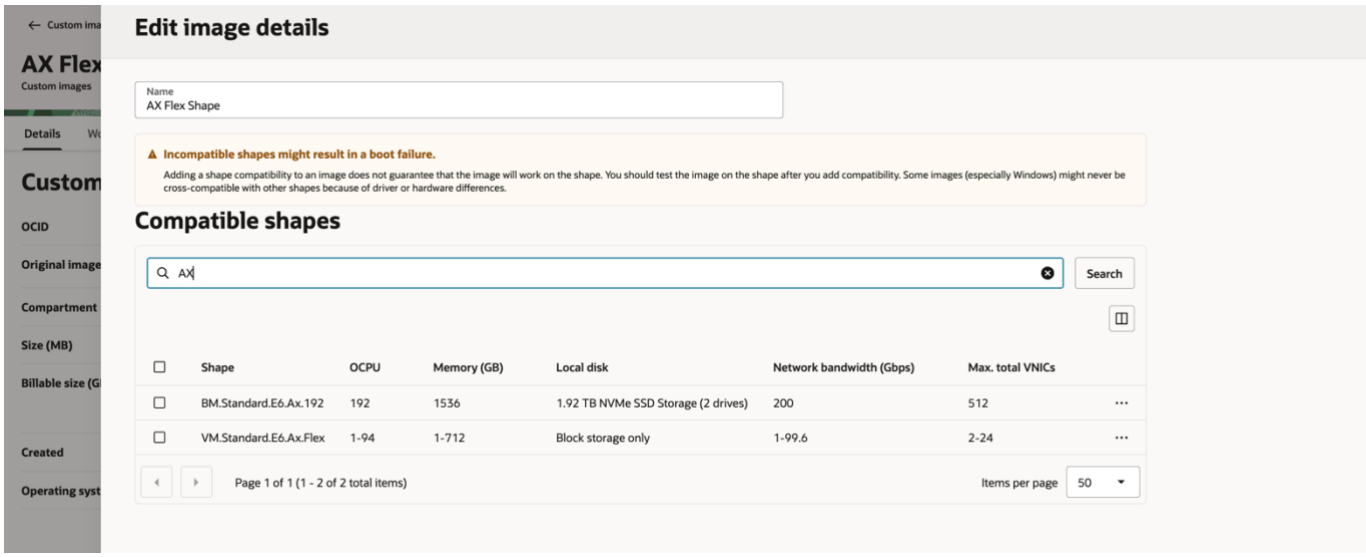


8. Go to Original Instance (The Instance you want to upgrade/change shape) → Actions → Create Custom Image and Provide a Name and Submit Create Custom Image. It will take some time to complete Create Custom Image process. **During the process, Instance will be in Shutdown state**. After completion, the instance will be reachable in Running state.



9. When Image is created, Go to **Compute → Custom Images** and note that Custom Image is available in the list.

10. Select **Custom Image → Actions → Edit Details** and Checkbox E6 Ax shapes listed and Save Changes.



11. Go to Instances and Select Instance → Storage → Replace Boot Volume and Select Custom Image Name from dropdown list.
    a. Check Preserve Boot Volume (Delete later after successful operation)
    b. Select Replace by Image & Select Custom Image Name
    c. Submit by clicking Replace.
    d. It will take some time to replace the Boot Volume disks and **Instance will not be reachable during the process**.

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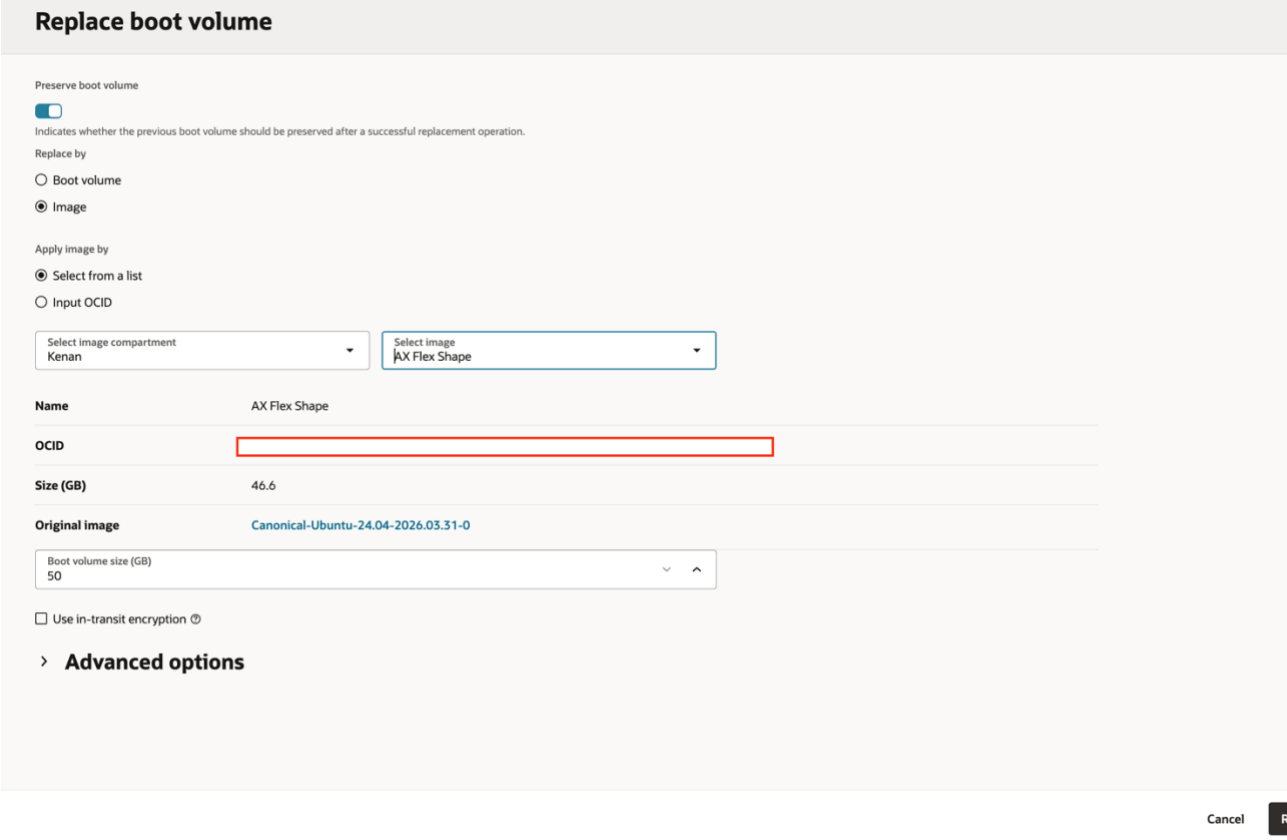

12. After Boot Volume replaced, the Instance will be in Running State and ready for Changing Current Shape to new Shape.
    a. Go to Edit Instance and Select New Shape (E6) and Modify OCPU and Memory as per requirement and Save Changes.
    b. Instance will be rebooted during the change and will be not reachable during that time.

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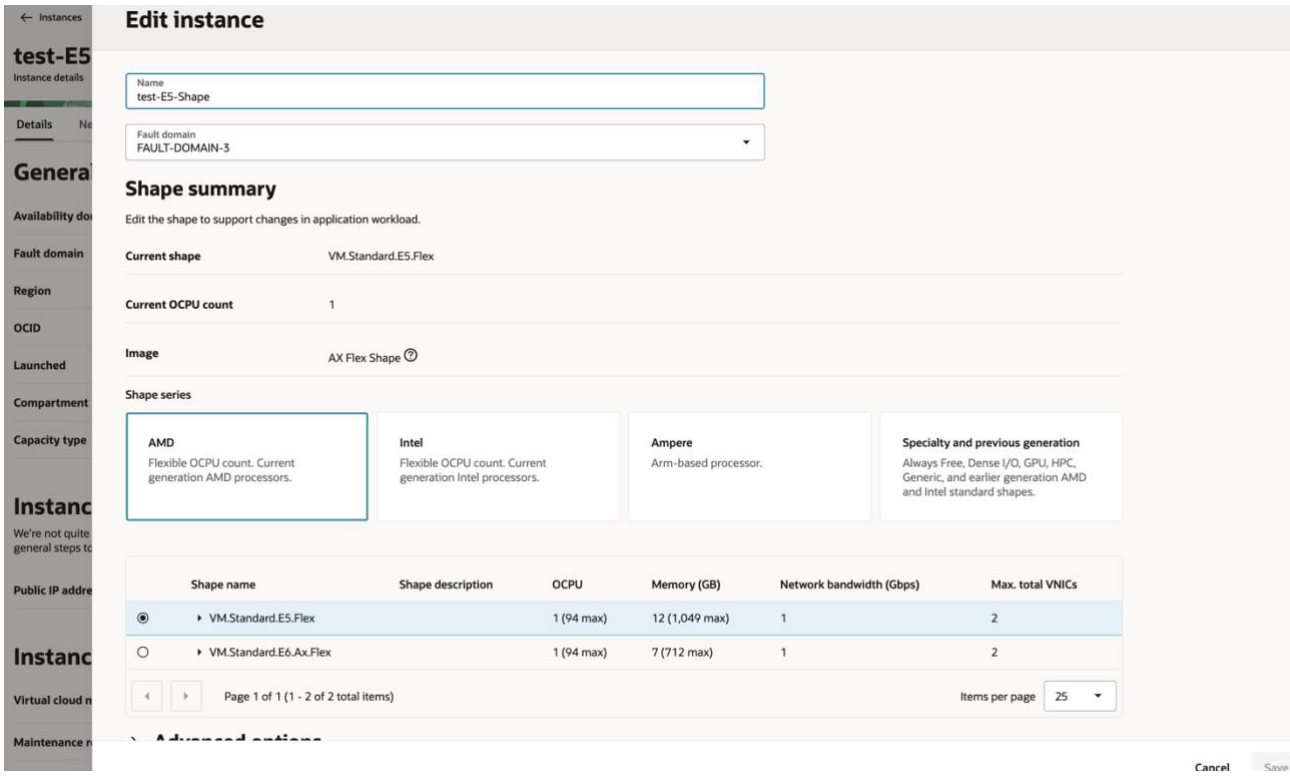
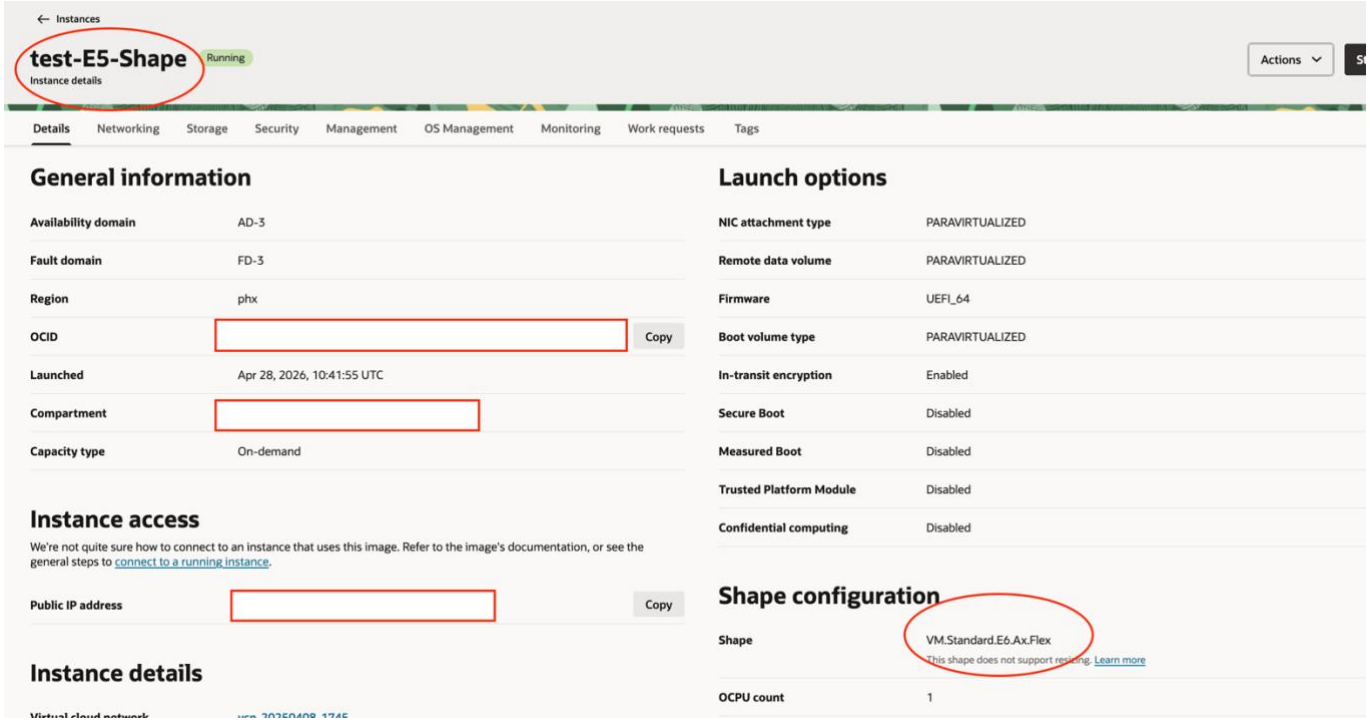


6.6 Post Activity Validation

- After instance is changed to new shape, you can connect via ssh with existing keys. As Boot Volume remains same, old keys will be persistent.



- **Private IP and MAC Address will be preserved in this method.**
- **OS validation**: CPU/memory recognized correctly; system logs are clean; services start successfully.
- **Application validation**: run health checks/smoke tests; confirm performance and connectivity.
- **(OPTIONAL)** To attach Additional Block Volumes, Go to OCI Console, Instance Details and Storage Tab, select Attach Block Volume and Select the required Volume with ISCSI attachment type.
    - After successful attachment, go to 3 dots next Block Volume name and copy ISCSI Connect Commands and paste to Linux Terminal.
    - As we have done during pre-checks, unmounting and arranging /etc/fstab entry, this time we need to mount disk with command such as # mount /dev/sdX and remove comment from /etc/fstab entry.
    - After mounting Block Volumes successfully, it is recommended to reboot Instance one time to see if it is booting up normally without issues.
- It is advised to check if OS and kernel updates are available.
- If the Instance is rebooting successfully, we can conclude the upgrade process of Instance.

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